Exploratory Data Analysis: European Soccer Matches, 2017–2018

A reproducible, detailed examination of team-match performance across five domestic leagues

Exploratory Data Analysis

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1 Executive overview

This report explores team-level match statistics from the 2017–18 season across five European domestic competitions: England, France, Germany, Italy, and Spain. The analytical unit is a **team-match**: each real match contributes two observations, one for each participating team.

The supplied analytical files contain 1,826 matches and 3,652 team-match records. The data include goals, goals conceded, shots, tagged shots on target, attempted and accurate passes, a derived passing-accuracy percentage, and a derived pass-share percentage. Supplementary files provide match-validation results, league-level home/away summaries, a numeric summary, a data dictionary, and a shot-type audit.

The central questions are:

1. How are goals, shots, passing volume, and passing accuracy distributed?
2. How do attacking and passing profiles differ across the five leagues?
3. Is there evidence of a home-team advantage in goals, shots, and passing measures?
4. How strongly are shots, passing volume, pass share, and goals associated?
5. Are there data-quality issues or measurement limitations that should shape interpretation?

Important measurement note: `pass_share_pct` is a proxy calculated from each team's share of the two teams' recorded passes. It is not official possession time. Likewise, `on_target_tagged` is based on an event tag and should be interpreted as a tagged-shot measure rather than assumed to be a perfect official shots-on-target statistic.

2 1. Research context and analytical design

2.1 1.1 Unit of analysis

The main dataset is structured as one row per team per match. A match between Team A and Team B should therefore have two rows sharing the same `match_id`. One row should identify the home team (`home = 1`) and the other the away team (`home = 0`).

This structure is useful for comparing team performance, but it has an important statistical consequence: the two observations from a match are not independent. They share the same match environment and opposing teams. Descriptive comparisons are appropriate, while inferential models should account for match-level clustering or use a match-level structure.

2.2 1.2 Variables

Variable	Meaning	Analytical role
`match_id`	Unique match identifier	Match grouping and validation
`match_date`	Match date (UTC)	Time trend and season coverage
`league`	Domestic competition	Grouping variable
`team_id`, `team_name`	Team identifier and label	Team-level grouping
`opponent_id`, `opponent_name`	Opponent identifier and label	Match context
`home`	1 = home, 0 = away	Venue comparison
`goals`	Goals scored by the team	Main attacking outcome
`goals_conceded`	Goals scored by opponent	Defensive outcome
`shots`	Recorded shot, free-kick-shot, or penalty events	Attacking volume
`on_target_tagged`	Events with accuracy tag 1801	Tagged shot accuracy proxy
`passes_attempted`	Events labeled Pass	Passing volume
`passes_accurate`	Passes tagged accurate	Accurate-pass count
`pass_accuracy_pct`	Accurate passes / attempted passes × 100	Passing accuracy

Variable	Meaning	Analytical role
`pass_share_pct`	Team passes / both teams' passes × 100	Passing share proxy

2.3 1.3 Analytical approach

The report combines data validation, descriptive statistics, distribution analysis, grouped comparisons, visual exploration, and association analysis. It does not treat correlations as causal effects. League differences may reflect playing styles, competition structures, team quality, event coverage, or other factors not measured here.

3 2. Setup and data import

The six CSV files are read directly from `C:/Users/Home/Downloads/New folder/`. If you move the CSV files, update `data_dir` in the setup chunk.

3.1 2.1 File inventory

```
file_inventory <- tibble(
  file = c(
    "soccer_team_match_2017_18.csv",
    "soccer_match_validation.csv",
    "soccer_home_away_summary.csv",
    "soccer_numeric_summary.csv",
    "soccer_data_dictionary.csv",
    "soccer_shot_type_audit.csv"
  ),
  purpose = c(
    "Primary team-match analytical dataset",
    "Match-level structural validation",
    "Precomputed league-by-venue summaries",
    "Precomputed numeric descriptive statistics",
    "Variable definitions and measurement notes",
    "Shot-event classification audit"
  )
)

knitr::kable(file_inventory, caption = "Supplied analytical files")
```

Table 1: Supplied analytical files

file	purpose
soccer_team_match_2017_18.csv	Primary team-match analytical dataset
soccer_match_validation.csv	Match-level structural validation
soccer_home_away_summary.csv	Precomputed league-by-venue summaries
soccer_numeric_summary.csv	Precomputed numeric descriptive statistics
soccer_data_dictionary.csv	Variable definitions and measurement notes
soccer_shot_type_audit.csv	Shot-event classification audit

3.2 2.2 Dataset dimensions and coverage

```
overview <- tibble(
  measure = c(
    "Team-match rows", "Unique matches", "Unique teams",
    "Leagues", "Distinct match dates", "First match date", "Last match date"
  ),
  value = c(
    nrow(team_match),
    n_distinct(team_match$match_id),
    n_distinct(team_match$team_id),
    n_distinct(team_match$league),
    n_distinct(team_match$match_date),
    as.character(min(team_match$match_date, na.rm = TRUE)),
    as.character(max(team_match$match_date, na.rm = TRUE))
  )
)

knitr::kable(overview, caption = "Overall data coverage")
```

Table 2: Overall data coverage

measure	value
Team-match rows	3652
Unique matches	1826
Unique teams	98
Leagues	5
Distinct match dates	178
First match date	2017-08-04
Last match date	2018-05-20

The data cover five leagues and use a team-match structure. A total of 3,652 team-match records corresponds to 1,826 matches if every match has exactly two team records. The validation section below checks whether this expected structure holds.

4 3. Data quality and validation

A strong exploratory analysis begins by checking whether the data structure matches the intended design. The checks below examine missingness, duplicate rows, identifier consistency, valid ranges, and the match-pair structure.

4.1 3.1 Column types and initial inspection

```
glimpse(team_match)
```

```
Rows: 3,652
Columns: 17
$ match_id          <dbl> 2499719, 2499719, 2499720, 2499720, 2499721, 2499721~
$ match_date        <date> 2017-08-11, 2017-08-11, 2017-08-12, 2017-08-12, 201~
$ league            <fct> England, England, England, England, England, England~
$ team_id           <dbl> 1609, 1631, 1651, 1625, 1610, 1646, 1628, 1673, 1623~
```

```
$ team_name         <chr> "Arsenal", "Leicester City", "Brighton & Hove Albion~
$ opponent_id       <dbl> 1631, 1609, 1625, 1651, 1646, 1610, 1673, 1628, 1639~
$ opponent_name     <chr> "Leicester City", "Arsenal", "Manchester City", "Bri~
$ home              <dbl> 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0~
$ goals             <dbl> 4, 3, 0, 2, 2, 3, 0, 3, 1, 0, 0, 0, 3, 3, 1, 0, 4, 0~
$ goals_conceded    <dbl> 3, 4, 2, 0, 3, 2, 3, 0, 0, 1, 0, 0, 3, 3, 0, 1, 0, 4~
$ passes_attempted  <dbl> 606, 230, 184, 754, 516, 296, 370, 273, 482, 267, 50~
$ passes_accurate   <dbl> 513, 166, 124, 682, 454, 245, 291, 194, 406, 205, 42~
$ shots             <dbl> 27, 7, 6, 14, 16, 10, 14, 8, 8, 10, 26, 4, 8, 13, 16~
$ on_target_tagged  <dbl> 10, 4, 2, 4, 6, 5, 4, 6, 3, 1, 2, 0, 4, 5, 6, 2, 5, ~
$ pass_share_pct    <dbl> 72.48804, 27.51196, 19.61620, 80.38380, 63.54680, 36~
$ pass_accuracy_pct <dbl> 84.65347, 72.17391, 67.39130, 90.45093, 87.98450, 82~
$ venue             <fct> Home, Away, Home, Away, Home, Away, Home, Away, Home~
```

4.2 3.2 Missing values

```
missingness <- team_match |>
  summarise(across(everything(), ~ sum(is.na(.x)))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "missing_n") |>
  mutate(
    missing_pct = 100 * missing_n / nrow(team_match)
  ) |>
  arrange(desc(missing_n))

knitr::kable(
  missingness,
  digits = 2,
  caption = "Missing observations by variable"
)
```

Table 3: Missing observations by variable

variable	missing_n	missing_pct
match_id	0	0
match_date	0	0
league	0	0
team_id	0	0
team_name	0	0
opponent_id	0	0
opponent_name	0	0
home	0	0
goals	0	0
goals_conceded	0	0
passes_attempted	0	0
passes_accurate	0	0
shots	0	0
on_target_tagged	0	0
pass_share_pct	0	0
pass_accuracy_pct	0	0
venue	0	0

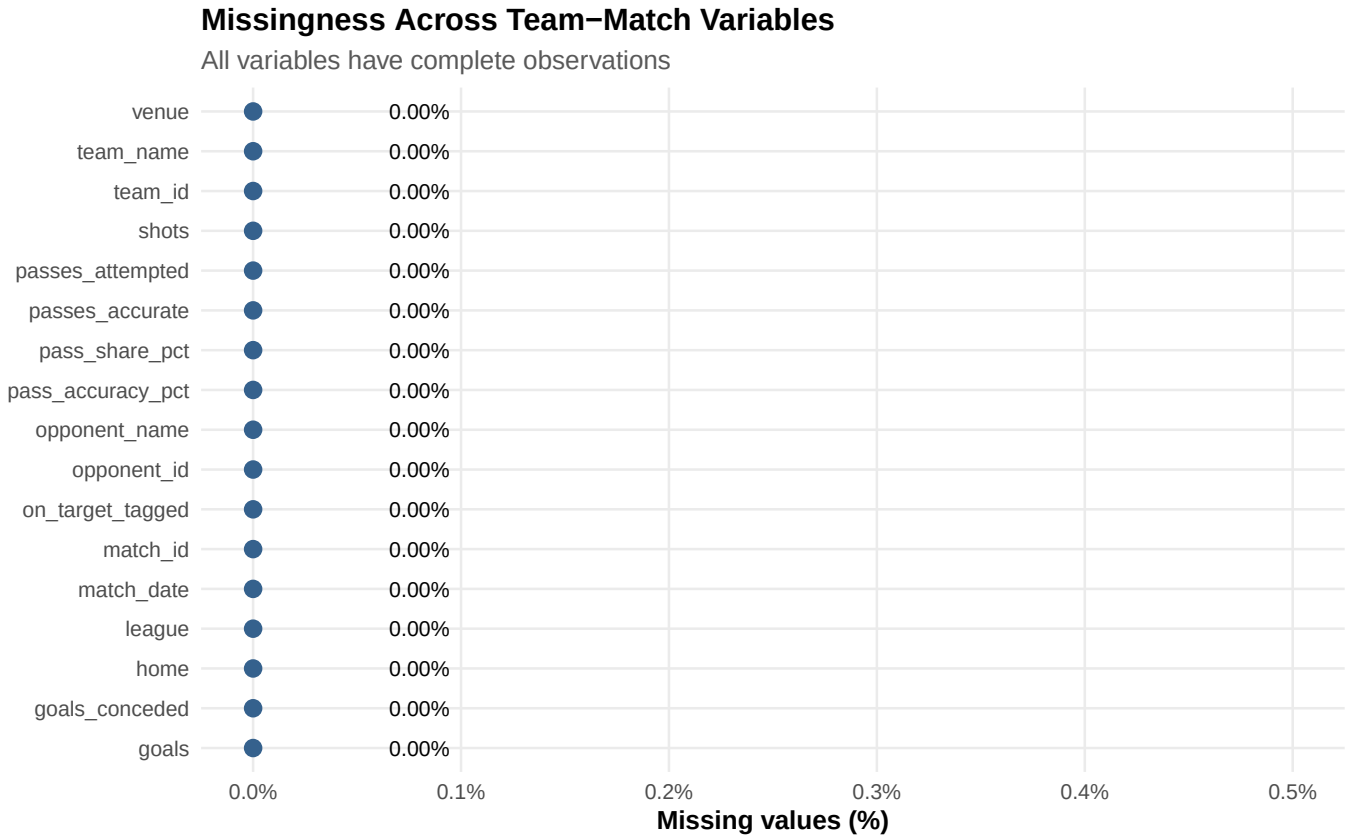
```
if (all(missingness$missing_n == 0)) {

  ggplot(
```

```
    missingness,
    aes(x = missing_pct, y = reorder(variable, missing_pct))
  ) +
    geom_point(color = "#35618D", size = 3) +
    geom_text(
      aes(label = sprintf("%.2f%%", missing_pct)),
      nudge_x = 0.08,
      size = 3.2
    ) +
    scale_x_continuous(
      limits = c(0, 0.5),
      breaks = seq(0, 0.5, by = 0.1),
      labels = scales::label_number(suffix = "%")
    ) +
    labs(
      title = "Missingness Across Team-Match Variables",
      subtitle = "All variables have complete observations",
      x = "Missing values (%)",
      y = NULL
    ) +
    theme_eda

} else {

  ggplot(
    missingness,
    aes(x = missing_pct, y = reorder(variable, missing_pct))
  ) +
    geom_col(fill = "#547C9B") +
    labs(
      title = "Missingness Across Team-Match Variables",
      x = "Missing values (%)",
      y = NULL
    ) +
    theme_eda
}
```



A variable with no missing values is complete in this supplied analytical extract, but completeness does not guarantee that the variable is measured without error. The tagged-event measures and derived percentages still require the interpretation caveats described above.

4.3 3.3 Duplicate records and identifier checks

```
duplicate_rows <- team_match |>
  count(across(everything()), name = "copies") |>
  filter(copies > 1)

duplicate_team_match <- team_match |>
  count(match_id, team_id, name = "records") |>
  filter(records > 1)

identifier_checks <- tibble(
  check = c(
    "Exact duplicate rows",
    "Repeated match_id + team_id combinations",
    "Missing match IDs",
    "Missing team IDs",
    "Missing opponent IDs",
    "Team and opponent IDs equal"
  ),
  problem_count = c(
    sum(duplicated(team_match)),
    nrow(duplicate_team_match),
    sum(is.na(team_match$match_id)),
    sum(is.na(team_match$team_id)),
```

```
    sum(is.na(team_match$opponent_id)),
    sum(team_match$team_id == team_match$opponent_id, na.rm = TRUE)
  )
)

knitr::kable(identifier_checks, caption = "Identifier and duplication checks")
```

Table 4: Identifier and duplication checks

check	problem_count
Exact duplicate rows	0
Repeated match_id + team_id combinations	0
Missing match IDs	0
Missing team IDs	0
Missing opponent IDs	0
Team and opponent IDs equal	0

4.4 3.4 Match-pair integrity

Each match should have two team records, two distinct teams, and exactly one home team. The supplied validation file allows us to inspect these conditions independently.

```
validation_summary <- validation |>
  summarise(
    matches_checked = n(),
    matches_with_two_rows = sum(rows == 2, na.rm = TRUE),
    matches_with_one_home = sum(home_count == 1, na.rm = TRUE),
    matches_with_two_teams = sum(distinct_teams == 2, na.rm = TRUE),
    matches_with_100_pass_share = sum(abs(pass_share_total - 100) < 1e-8, na.rm = TRUE),
    matches_with_missing_pass_share = sum(pass_share_missing, na.rm = TRUE)
  )

knitr::kable(validation_summary, caption = "Summary of supplied match validation")
```

Table 5: Summary of supplied match validation

matches_checked	matches_with_two_rows	matches_with_one_home	matches_with_two_teams	matches_with_100_pass_share	matches_with_missing_pass_share
1826	1826	1826	1826	1826	0

```
validation %>%
  summarise(
    across(
      c(rows, home_count, distinct_teams, pass_share_total),
      list(min = ~ min(.x, na.rm = TRUE),
           median = ~ median(.x, na.rm = TRUE),
           max = ~ max(.x, na.rm = TRUE))
    )
  ) |>
  pivot_longer(everything(), names_to = "metric", values_to = "value") |>
  knitr::kable(digits = 2, caption = "Distribution of match-validation fields")
```

Table 6: Distribution of match-validation fields

metric	value
rows_min	2
rows_median	2
rows_max	2
home_count_min	1
home_count_median	1
home_count_max	1
distinct_teams_min	2
distinct_teams_median	2
distinct_teams_max	2
pass_share_total_min	100
pass_share_total_median	100
pass_share_total_max	100

The supplied validation summary reports 1,826 matches. The expected pattern is two rows per match, one home team, two distinct teams, and a combined pass share of 100%. These checks support the internal consistency of the team-match structure and the pass-share calculation.

4.5 3.5 Range and logical consistency checks

```
range_checks <- tibble(
  check = c(
    "home is either 0 or 1",
    "goals are non-negative",
    "goals_conceded are non-negative",
    "shots are non-negative",
    "on_target_tagged is non-negative",
    "passes_accurate does not exceed passes_attempted",
    "pass_accuracy_pct is between 0 and 100",
    "pass_share_pct is between 0 and 100",
    "goals scored equals opponent goals conceded within match"
  ),
  failures = c(
    sum(!team_match$home %in% c(0, 1), na.rm = TRUE),
    sum(team_match$goals < 0, na.rm = TRUE),
    sum(team_match$goals_conceded < 0, na.rm = TRUE),
    sum(team_match$shots < 0, na.rm = TRUE),
    sum(team_match$on_target_tagged < 0, na.rm = TRUE),
    sum(team_match$passes_accurate > team_match$passes_attempted, na.rm = TRUE),
    sum(team_match$pass_accuracy_pct < 0 | team_match$pass_accuracy_pct > 100, na.rm = TRUE),
    sum(team_match$pass_share_pct < 0 | team_match$pass_share_pct > 100, na.rm = TRUE),
    NA_integer_
  )
)

match_score_check <- team_match |>
  select(match_id, team_id, opponent_id, goals, goals_conceded) |>
  inner_join(
    team_match |>
      transmute(
        match_id,
        opponent_id = team_id,
```

```
        team_id = opponent_id,
        opponent_goals = goals,
        opponent_conceded = goals_conceded
      ),
    by = c("match_id", "team_id", "opponent_id")
  ) |>
  summarise(
    mismatched_score_pairs = sum(
      goals != opponent_goals | goals_conceded != opponent_conceded,
      na.rm = TRUE
    )
  )

range_checks$failures[9] <- match_score_check$mismatched_score_pairs

knitr::kable(range_checks, caption = "Logical and range checks")
```

Table 7: Logical and range checks

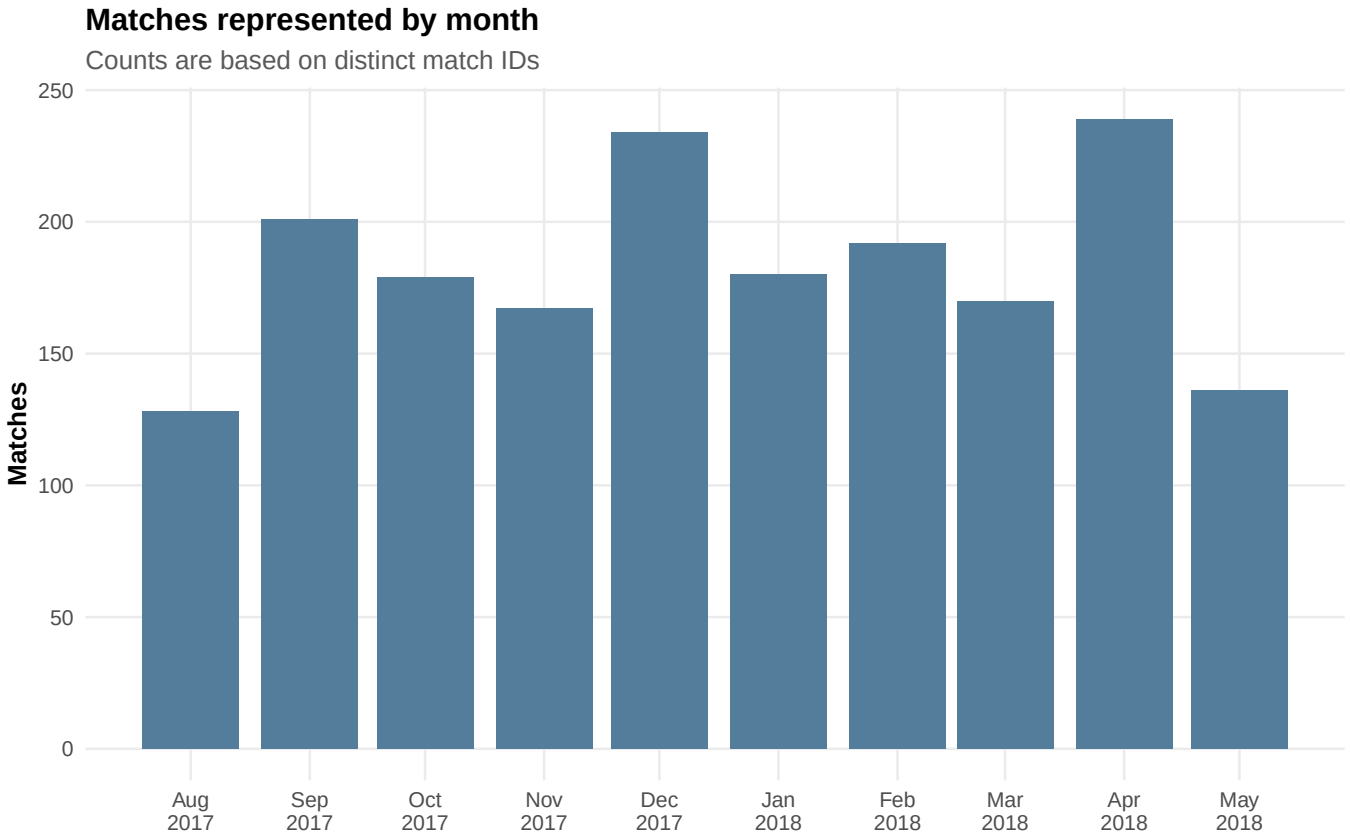
check	failures
home is either 0 or 1	0
goals are non-negative	0
goals_conceded are non-negative	0
shots are non-negative	0
on_target_tagged is non-negative	0
passes_accurate does not exceed passes_attempted	0
pass_accuracy_pct is between 0 and 100	0
pass_share_pct is between 0 and 100	0
goals scored equals opponent goals conceded within match	0

The final check compares each team's score with the opponent's corresponding record. A zero count indicates that the paired team rows agree on the match score.

4.6 3.6 Date coverage and match counts over time

```
matches_by_month <- team_match |>
  distinct(match_id, match_date, league) |>
  mutate(month = floor_date(match_date, "month")) |>
  count(month, name = "matches")

ggplot(matches_by_month, aes(month, matches)) +
  geom_col(fill = "#547C9B", width = 25) +
  scale_x_date(date_breaks = "1 month", date_labels = "%b\n%Y") +
  labs(
    title = "Matches represented by month",
    subtitle = "Counts are based on distinct match IDs",
    x = NULL, y = "Matches"
  ) +
  theme_eda
```



Season schedules are not evenly distributed across calendar months. Month-to-month changes in match counts should therefore be interpreted as schedule coverage, not automatically as changes in league activity or performance.

5 4. Descriptive statistics

5.1 4.1 Overall numeric summary

```
overall_summary <- team_match |>
  summarise(
    across(
      all_of(numeric_vars),
      list(
        n = ~ sum(!is.na(.x)),
        mean = ~ mean(.x, na.rm = TRUE),
        sd = ~ sd(.x, na.rm = TRUE),
        median = ~ median(.x, na.rm = TRUE),
        q1 = ~ quantile(.x, 0.25, na.rm = TRUE),
        q3 = ~ quantile(.x, 0.75, na.rm = TRUE),
        min = ~ min(.x, na.rm = TRUE),
        max = ~ max(.x, na.rm = TRUE)
      ),
      .names = "{.col}_{.fn}"
    )
  ) |>
  pivot_longer(
    everything(),
    names_to = c("variable", ".value"),
```

```
    names_pattern = "(.+)_([^_]+)$"
  )

knitr::kable(overall_summary, digits = 2, caption = "Expanded descriptive statistics")
```

Table 8: Expanded descriptive statistics

variable	n	mean	sd	median	q1	q3	min	max
goals	3652	1.35	1.26	1.00	0.00	2.00	0.00	8.00
goals_conceded	3652	1.35	1.26	1.00	0.00	2.00	0.00	8.00
shots	3652	11.79	4.96	11.00	8.00	15.00	0.00	39.00
on_target_tagged	3652	4.25	2.50	4.00	2.00	6.00	0.00	15.00
passes_attempted	3652	428.63	126.63	411.50	339.00	507.00	121.00	1008.00
passes_accurate	3652	355.00	124.07	337.00	267.00	428.25	86.00	943.00
pass_accuracy_pct	3652	81.55	5.45	82.11	78.18	85.46	58.21	93.71
pass_share_pct	3652	50.00	12.48	50.00	40.97	59.03	15.20	84.80

The mean and median are both useful. For count variables such as goals and shots, a small number of unusually high values can pull the mean upward. The median and interquartile range provide a more robust description of a typical team-match.

5.2 4.2 Supplied numeric summary

```
knitr::kable(
  numeric_summary,
  digits = 2,
  caption = "Provided numeric summary: minimum, median, mean, 95th percentile, and maximum"
)
```

Table 9: Provided numeric summary: minimum, median, mean, 95th percentile, and maximum

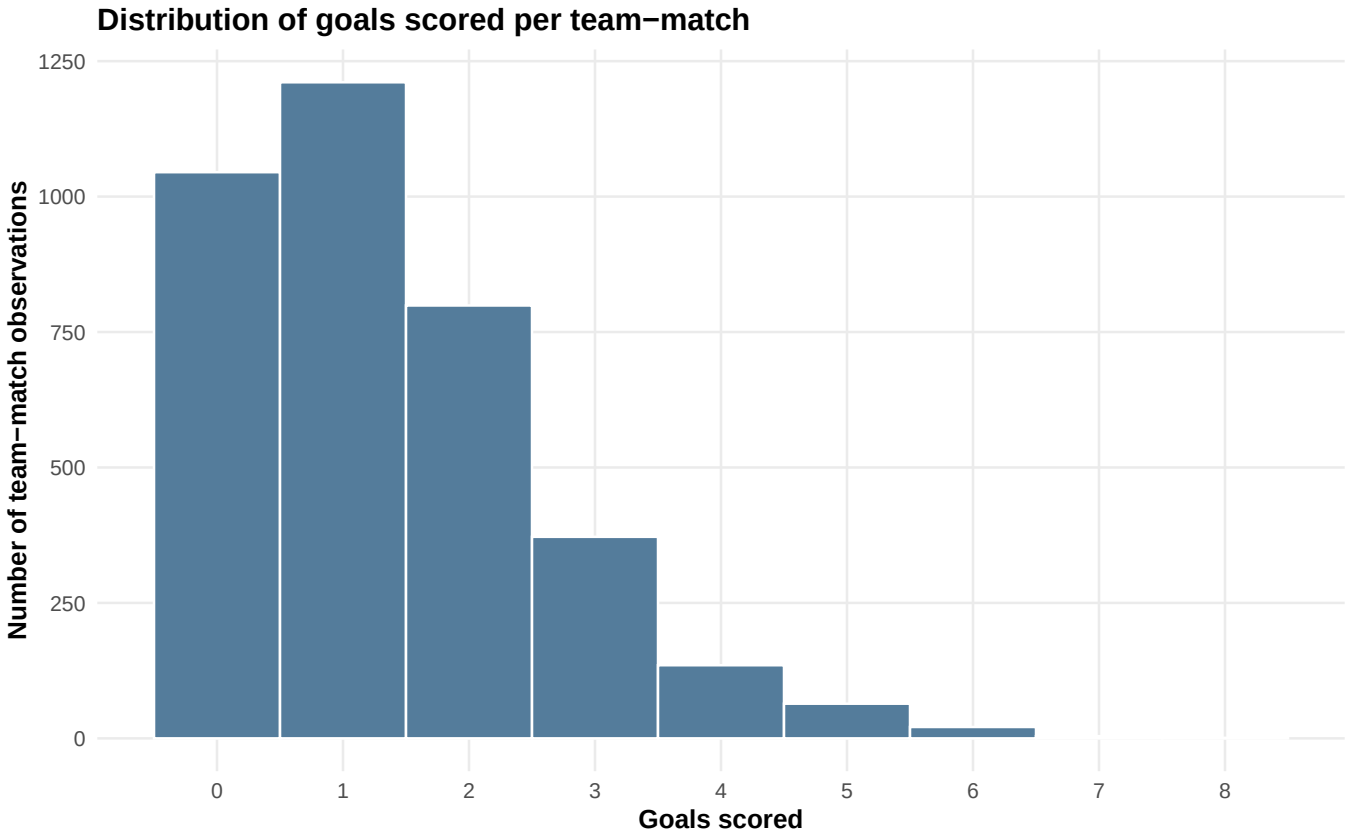
metric	n	minimum	median	mean	p95	maximum
goals	3652	0.00	1.00	1.35	4.00	8.00
on_target_tagged	3652	0.00	4.00	4.25	9.00	15.00
pass_accuracy_pct	3652	58.21	82.11	81.55	89.60	93.71
pass_share_pct	3652	15.20	50.00	50.00	70.26	84.80
passes_attempted	3652	121.00	411.50	428.63	660.45	1008.00
shots	3652	0.00	11.00	11.79	21.00	39.00

The supplied summary indicates an average of approximately 1.35 goals per team-match, 11.79 shots, and 4.25 tagged shots on target. Average passing accuracy is approximately 81.55%. The average pass share is exactly 50% by construction because each match's two team shares sum to 100%.

5.3 4.3 Distribution of goals

```
ggplot(team_match, aes(goals)) +
  geom_histogram(binwidth = 1, boundary = -0.5,
                 fill = "#547C9B", color = "white") +
  scale_x_continuous(breaks = 0:max(team_match$goals, na.rm = TRUE)) +
```

```
  labs(
    title = "Distribution of goals scored per team-match",
    x = "Goals scored", y = "Number of team-match observations"
  ) +
  theme_eda
```




```
goals_frequency <- team_match |>
  count(goals, name = "team_matches") |>
  mutate(percent = 100 * team_matches / sum(team_matches))

knitr::kable(goals_frequency, digits = 2,
             caption = "Frequency distribution of goals scored")
```

Table 10: Frequency distribution of goals scored

goals	team_matches	percent
0	1045	28.61
1	1211	33.16
2	799	21.88
3	372	10.19
4	135	3.70
5	64	1.75
6	21	0.58
7	4	0.11
8	1	0.03

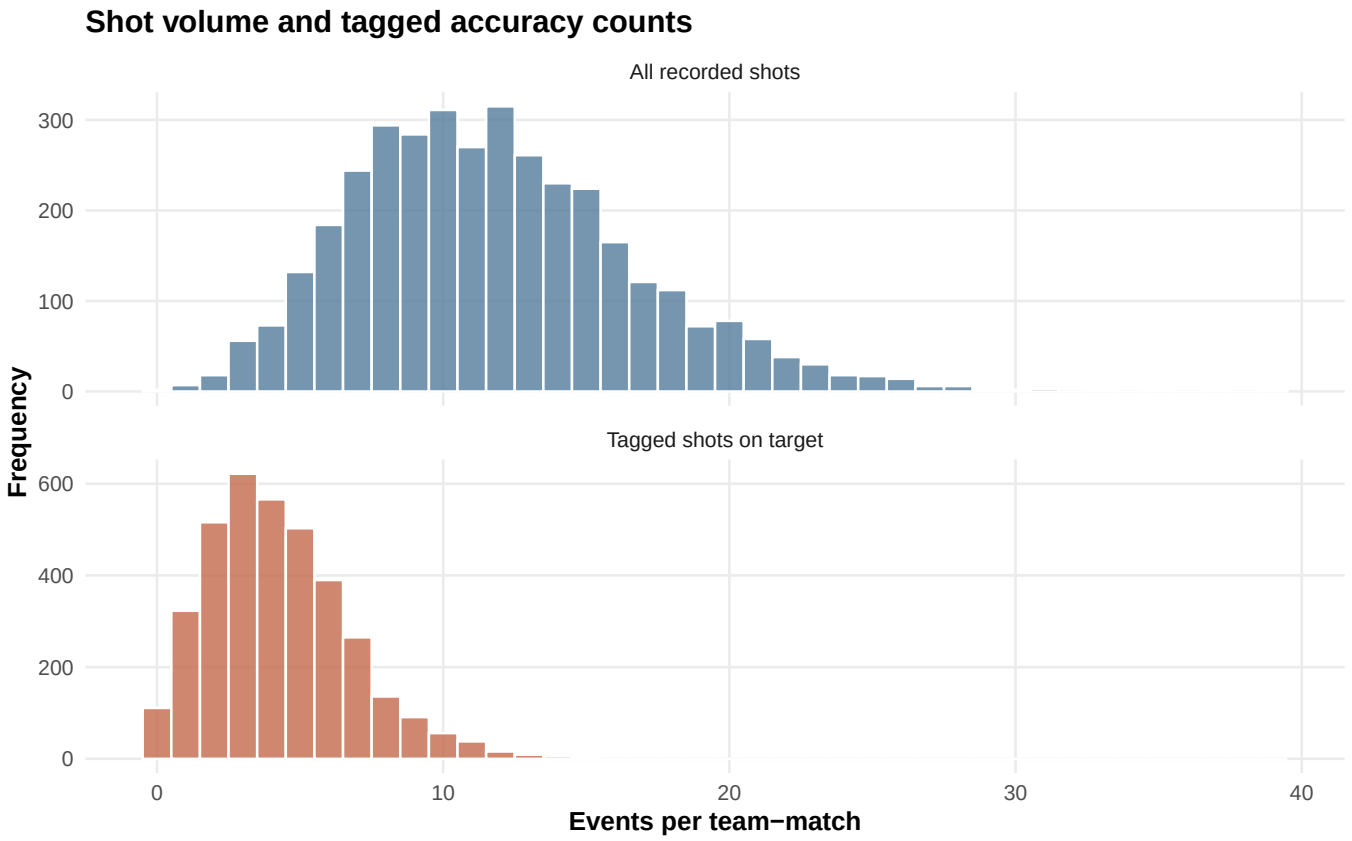
Goals are a discrete, low-count outcome. A substantial concentration at zero, one, and two goals is typical of football

score data. This distribution is not well described by a normal curve; count-based models may be more appropriate if the analysis moves beyond descriptive exploration.

5.4 4.4 Distribution of shots and tagged shots on target

```
shot_long <- team_match |>
  select(shots, on_target_tagged) |>
  pivot_longer(everything(), names_to = "measure", values_to = "count") |>
  mutate(
    measure = recode(
      measure,
      shots = "All recorded shots",
      on_target_tagged = "Tagged shots on target"
    )
  )

ggplot(shot_long, aes(count, fill = measure)) +
  geom_histogram(binwidth = 1, boundary = -0.5, alpha = 0.8,
                 color = "white") +
  facet_wrap(~ measure, scales = "free_y", ncol = 1) +
  scale_fill_manual(values = c("All recorded shots" = "#547C9B",
                               "Tagged shots on target" = "#C36B4B")) +
  labs(
    title = "Shot volume and tagged accuracy counts",
    x = "Events per team-match", y = "Frequency"
  ) +
  theme_eda +
  theme(legend.position = "none")
```



```
shot_efficiency <- team_match |>
  mutate(
    tagged_on_target_rate = if_else(shots > 0,
                                    100 * on_target_tagged / shots,
                                    NA_real_)
  ) |>
  summarise(
    team_matches = n(),
    total_shots = sum(shots, na.rm = TRUE),
    total_tagged_on_target = sum(on_target_tagged, na.rm = TRUE),
    aggregate_tagged_rate = 100 * total_tagged_on_target / total_shots,
    median_team_match_rate = median(tagged_on_target_rate, na.rm = TRUE),
    zero_shot_matches = sum(shots == 0, na.rm = TRUE),
    zero_shot_on_target_mismatches = sum(shots == 0 & on_target_tagged > 0, na.rm = TRUE)
  )

knitr::kable(shot_efficiency, digits = 2,
             caption = "Aggregate and team-match shot-tag summaries")
```

Table 11: Aggregate and team-match shot-tag summaries

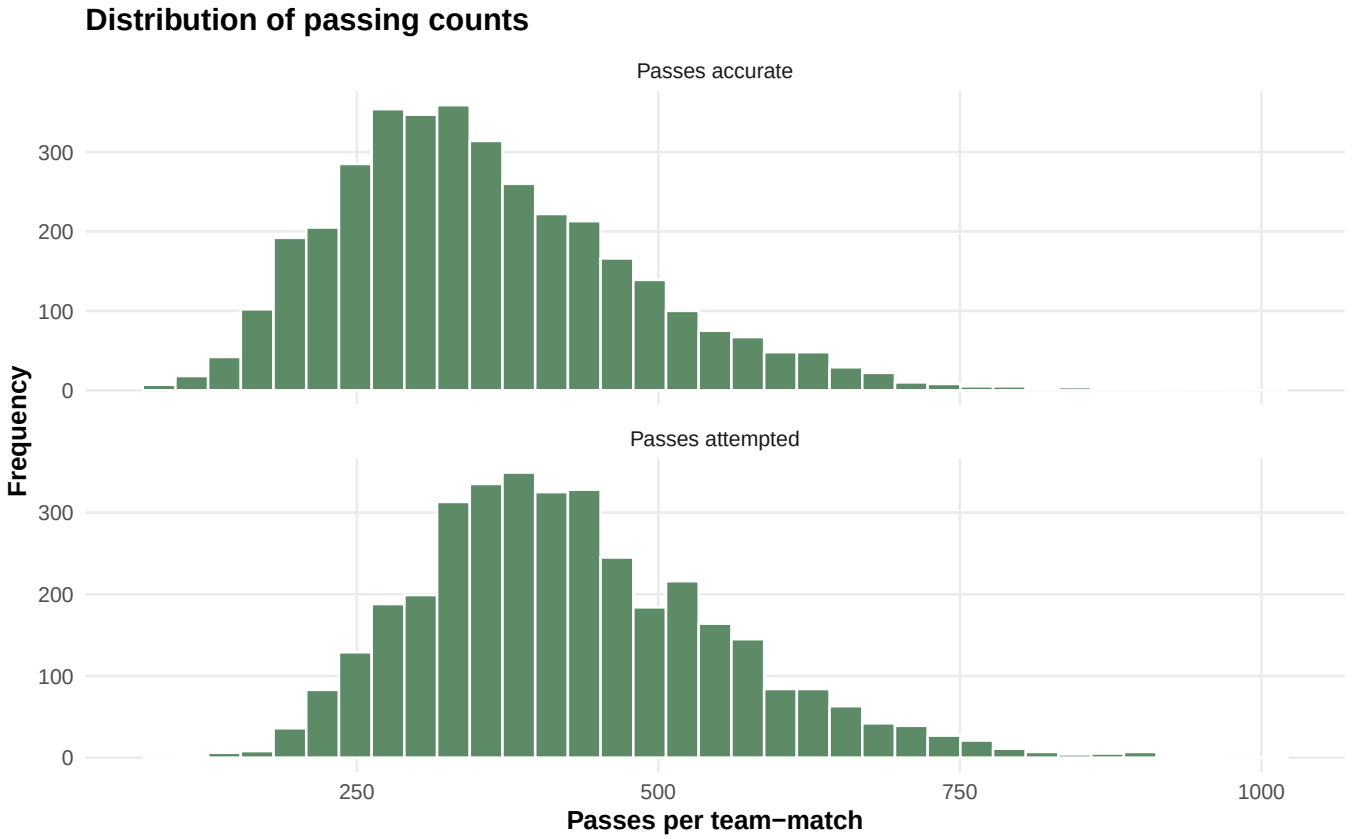
team_matches	total_shots	total_tagged_on_target	aggregate_tagged_rate	median_team_match_rate	zero_shot_matches	zero_shot_on_target_mismatches
3652	43040	15533	36.09	35.71	2	0

The aggregate tagged rate weights high-shot matches more heavily than low-shot matches. The median team-match rate instead describes the typical rate among team-matches with at least one recorded shot. These are different estimands and should not be treated as interchangeable.

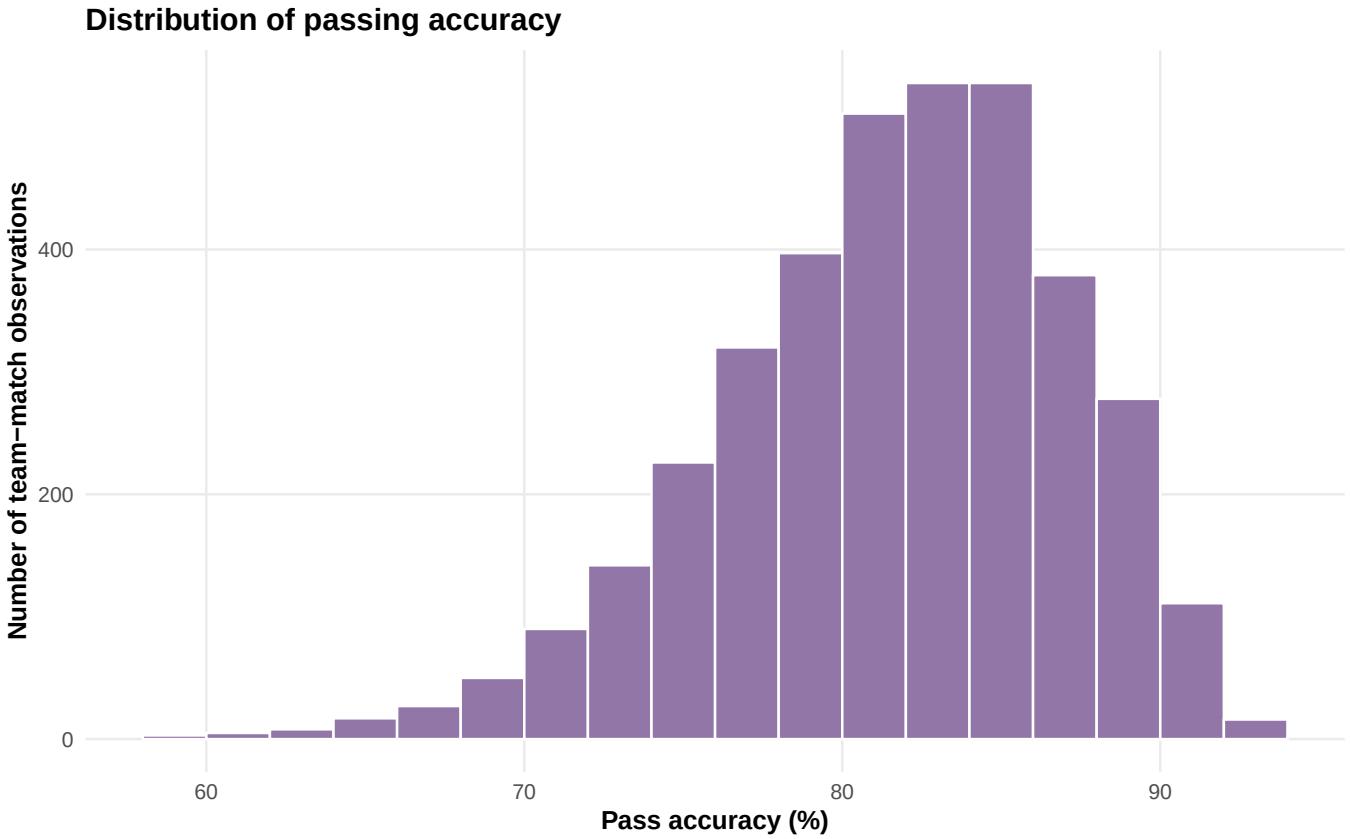
5.5 4.5 Passing volume and accuracy

```
pass_long <- team_match |>
  select(passes_attempted, passes_accurate) |>
  pivot_longer(everything(), names_to = "measure", values_to = "passes") |>
  mutate(
    measure = recode(
      measure,
      passes_attempted = "Passes attempted",
      passes_accurate = "Passes accurate"
    )
  )

ggplot(pass_long, aes(passes)) +
  geom_histogram(bins = 35, fill = "#5D8B68", color = "white") +
  facet_wrap(~ measure, scales = "free_y", ncol = 1) +
  labs(
    title = "Distribution of passing counts",
    x = "Passes per team-match", y = "Frequency"
  ) +
  theme_eda
```




```
ggplot(team_match, aes(pass_accuracy_pct)) +
  geom_histogram(binwidth = 2, boundary = 0,
                 fill = "#9276A8", color = "white") +
  labs(
    title = "Distribution of passing accuracy",
    x = "Pass accuracy (%)", y = "Number of team-match observations"
  ) +
  theme_eda
```



Passing accuracy is a ratio, not a count. Two teams can have the same accuracy percentage while attempting very different numbers of passes. Any comparison of passing accuracy should therefore be considered alongside passing volume.

6 5. League-level comparisons

6.1 5.1 Number of matches and teams by league

```
league_coverage <- team_match |>
  group_by(league) |>
  summarise(
    team_matches = n(),
    matches = n_distinct(match_id),
    teams = n_distinct(team_id),
    first_date = min(match_date, na.rm = TRUE),
    last_date = max(match_date, na.rm = TRUE),
    .groups = "drop"
  )

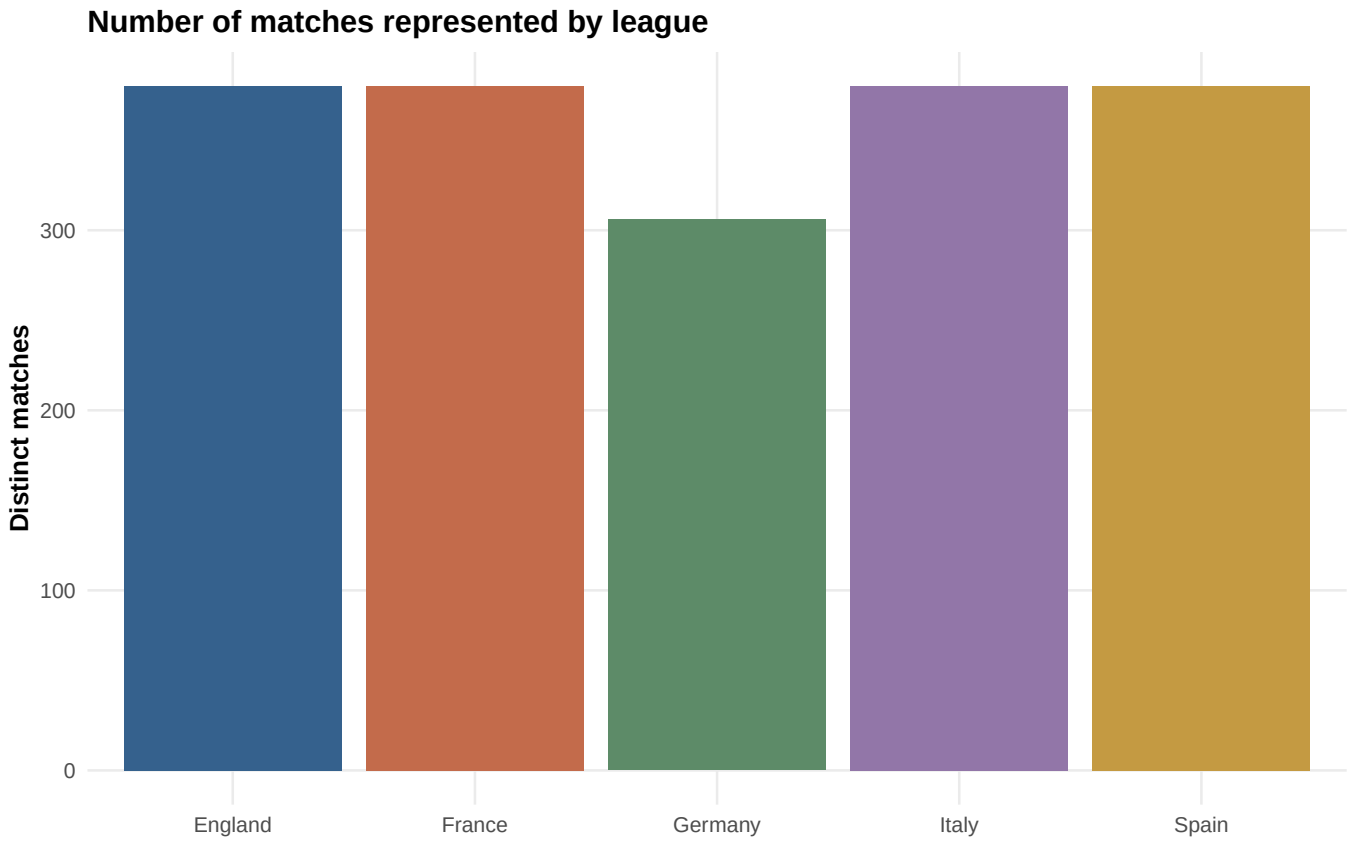
knitr::kable(league_coverage, caption = "League coverage and sample sizes")
```

Table 12: League coverage and sample sizes

league	team_matches	matches	teams	first_date	last_date
England	760	380	20	2017-08-11	2018-05-13
France	760	380	20	2017-08-04	2018-05-19

league	team_matches	matches	teams	first_date	last_date
Germany	612	306	18	2017-08-18	2018-05-12
Italy	760	380	20	2017-08-19	2018-05-20
Spain	760	380	20	2017-08-18	2018-05-20

```
ggplot(league_coverage, aes(league, matches, fill = league)) +
  geom_col(show.legend = FALSE) +
  scale_fill_manual(values = league_palette) +
  labs(
    title = "Number of matches represented by league",
    x = NULL, y = "Distinct matches"
  ) +
  theme_eda
```



The league samples are not identical in size: Germany has 306 matches, while each of the other four leagues has 380. When comparing totals, account for these different sample sizes. Per-team-match averages are more suitable for descriptive comparisons.

6.2 5.2 League-level attacking and passing profile

```
league_summary <- team_match |>
  group_by(league) |>
  summarise(
    team_matches = n(),
    avg_goals = mean(goals, na.rm = TRUE),
    median_goals = median(goals, na.rm = TRUE),
```

```
    avg_goals_conceded = mean(goals_conceded, na.rm = TRUE),
    avg_shots = mean(shots, na.rm = TRUE),
    avg_tagged_on_target = mean(on_target_tagged, na.rm = TRUE),
    avg_passes = mean(passes_attempted, na.rm = TRUE),
    avg_pass_accuracy = mean(pass_accuracy_pct, na.rm = TRUE),
    avg_pass_share = mean(pass_share_pct, na.rm = TRUE),
    .groups = "drop"
  )

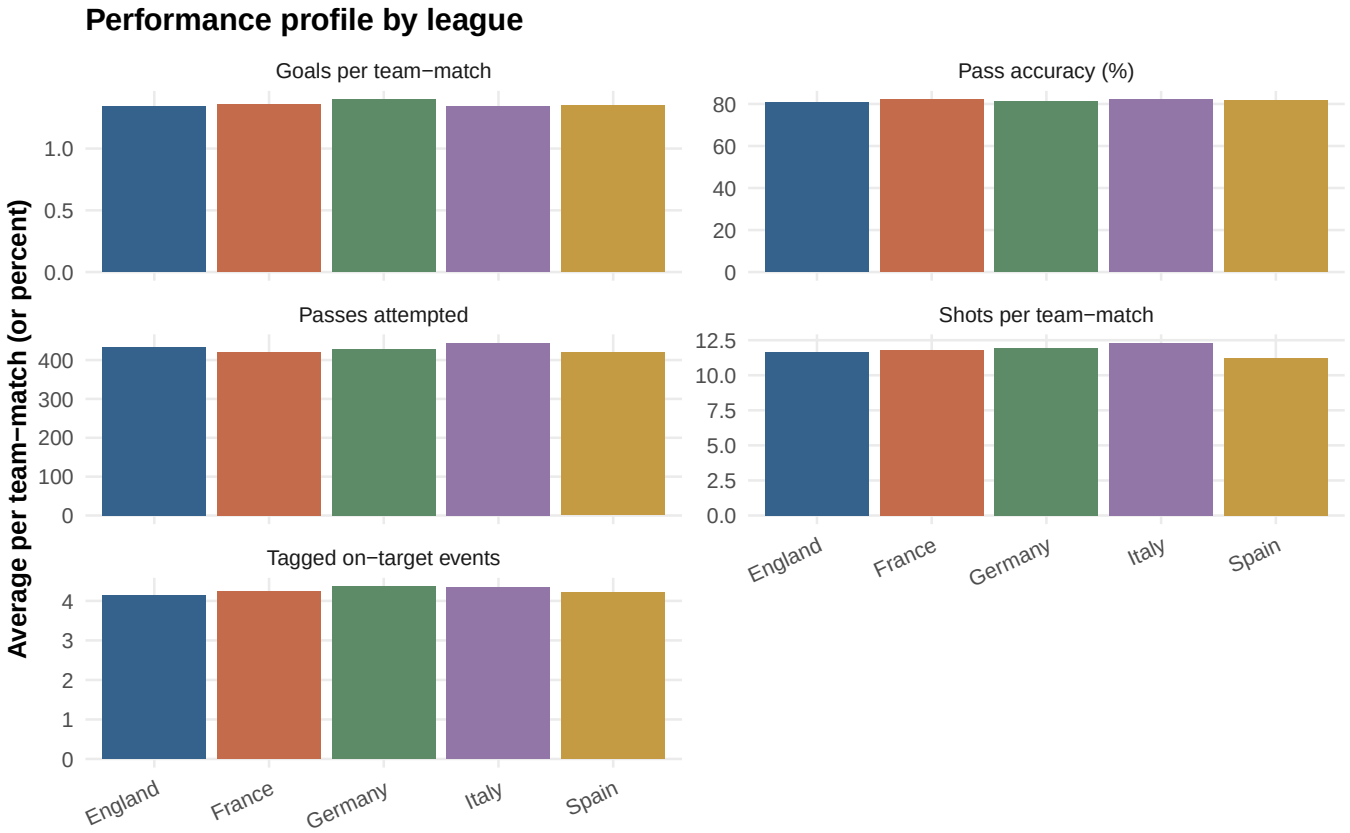
knitr::kable(league_summary, digits = 2,
             caption = "League-level team-match performance averages")
```

Table 13: League-level team-match performance averages

league	team_matches	avg_goals	median_goals	avg_goals_conceded	avg_shots	avg_tagged_on_target	avg_passes	avg_pass_accuracy	avg_pass_share
England	760	1.34	1	1.34	11.69	4.13	432.44	80.89	50
France	760	1.36	1	1.36	11.81	4.24	420.00	82.04	50
Germany	612	1.40	1	1.40	11.91	4.37	427.23	80.98	50
Italy	760	1.34	1	1.34	12.30	4.34	443.84	82.10	50
Spain	760	1.35	1	1.35	11.24	4.20	419.37	81.62	50

```
league_metric_long <- league_summary |>
  select(league, avg_goals, avg_shots, avg_tagged_on_target,
         avg_passes, avg_pass_accuracy) |>
  pivot_longer(-league, names_to = "metric", values_to = "value") |>
  mutate(
    metric = recode(
      metric,
      avg_goals = "Goals per team-match",
      avg_shots = "Shots per team-match",
      avg_tagged_on_target = "Tagged on-target events",
      avg_passes = "Passes attempted",
      avg_pass_accuracy = "Pass accuracy (%)"
    )
  )

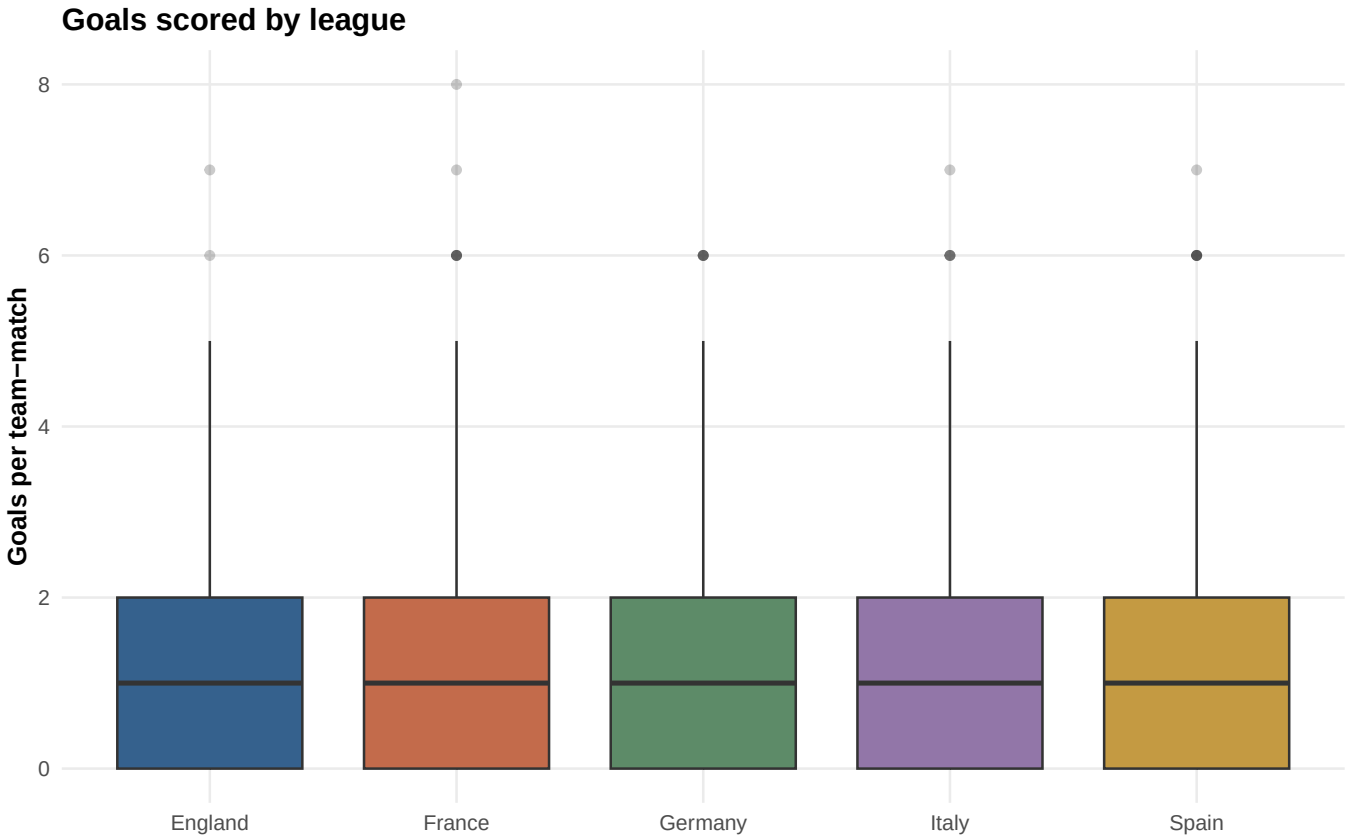
ggplot(league_metric_long, aes(league, value, fill = league)) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ metric, scales = "free_y", ncol = 2) +
  scale_fill_manual(values = league_palette) +
  labs(
    title = "Performance profile by league",
    x = NULL, y = "Average per team-match (or percent)"
  ) +
  theme_eda +
  theme(axis.text.x = element_text(angle = 25, hjust = 1))
```



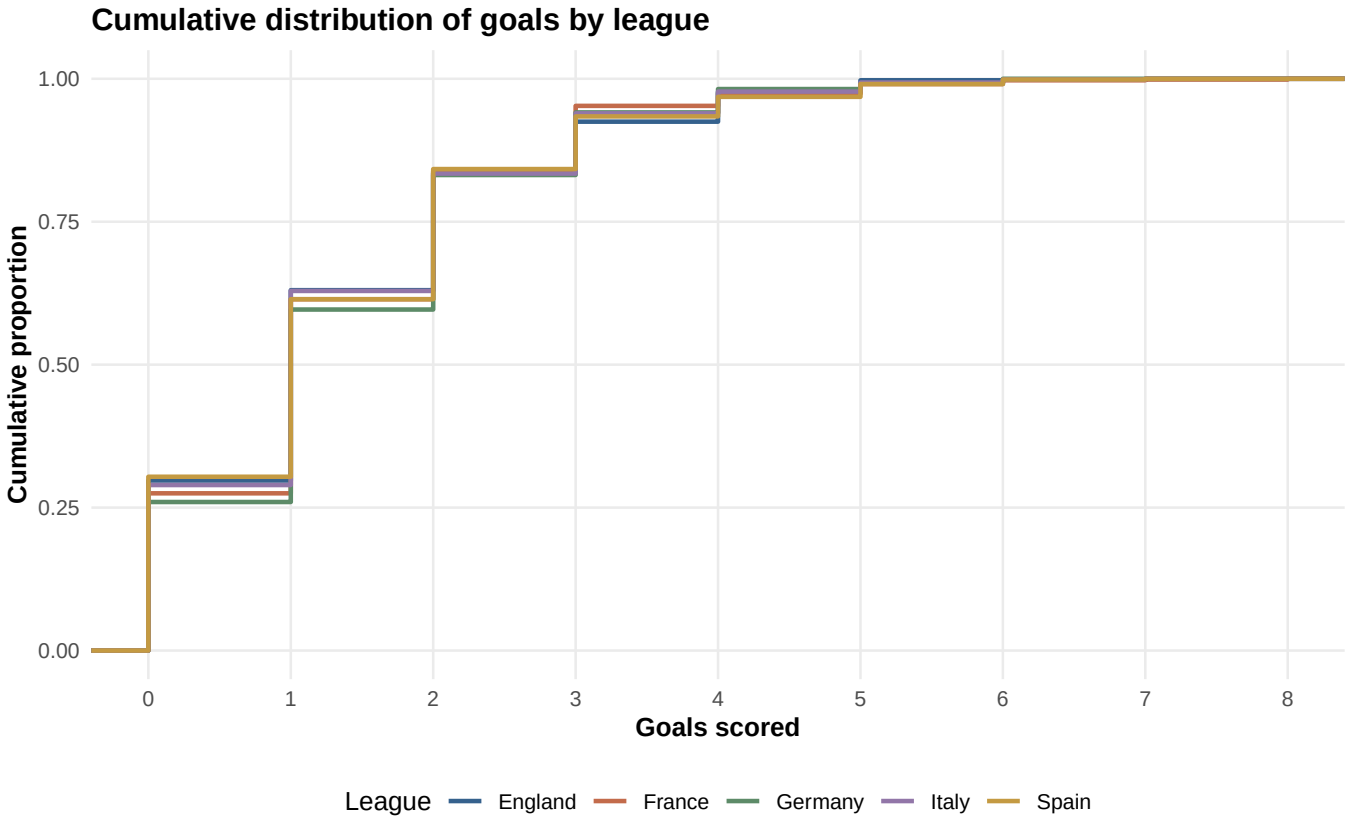
The facets use independent y-axis scales because the measures have different units. Compare leagues within each panel, not the visual heights across different panels.

6.3 5.3 Goal distributions by league

```
ggplot(team_match, aes(league, goals, fill = league)) +
  geom_boxplot(show.legend = FALSE, outlier.alpha = 0.25) +
  scale_fill_manual(values = league_palette) +
  labs(
    title = "Goals scored by league",
    x = NULL, y = "Goals per team-match"
  ) +
  theme_eda
```




```
ggplot(team_match, aes(goals, color = league)) +
  stat_ecdf(linewidth = 0.9) +
  scale_color_manual(values = league_palette) +
  scale_x_continuous(breaks = 0:8) +
  labs(
    title = "Cumulative distribution of goals by league",
    x = "Goals scored", y = "Cumulative proportion",
    color = "League"
  ) +
  theme_eda
```



The boxplot emphasizes medians and spread, while the empirical cumulative distribution compares the full distributions. Differences in means can conceal whether a league has more low-scoring matches, a heavier high-scoring tail, or both.

6.4 5.4 Team-level variation within leagues

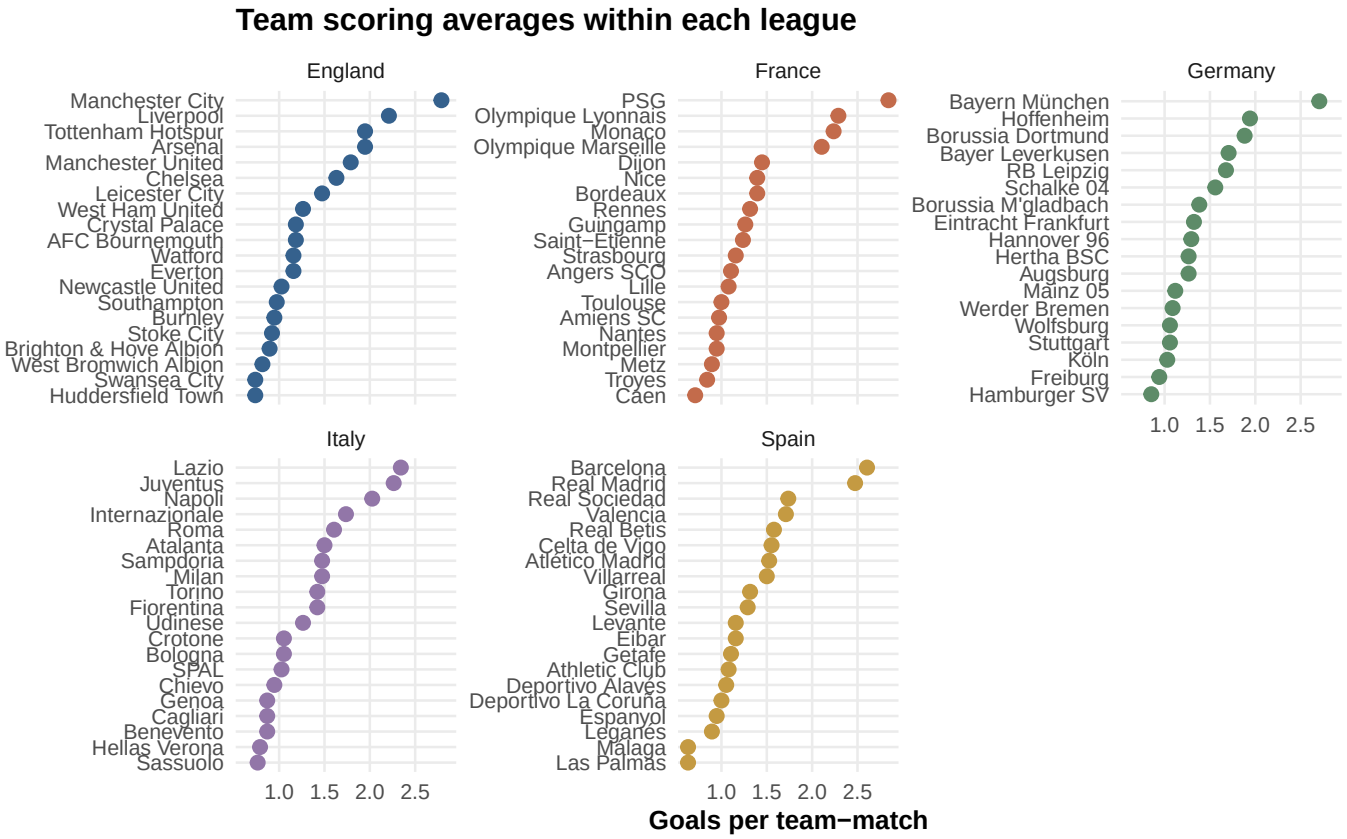
```
team_summary <- team_match |>
  group_by(league, team_id, team_name) |>
  summarise(
    team_matches = n(),
    goals_per_match = mean(goals, na.rm = TRUE),
    goals_conceded_per_match = mean(goals_conceded, na.rm = TRUE),
    goal_difference_per_match = mean(goals - goals_conceded, na.rm = TRUE),
    shots_per_match = mean(shots, na.rm = TRUE),
    pass_accuracy = mean(pass_accuracy_pct, na.rm = TRUE),
    .groups = "drop"
  )

team_summary |>
  group_by(league) |>
  slice_max(goals_per_match, n = 5, with_ties = FALSE) |>
  arrange(league, desc(goals_per_match)) |>
  knitr::kable(digits = 2, caption = "Five highest-scoring teams within each league")
```

Table 14: Five highest-scoring teams within each league

league	team_id	team_name	team_matches	goals_per_match	goals_conceded_per_match	goal_difference_per_match	shots_per_match	pass_accuracy
England	1625	Manchester City	38	2.79	0.71	2.08	16.92	89.62
England	1612	Liverpool	38	2.21	1.00	1.21	16.45	85.25
England	1609	Arsenal	38	1.95	1.34	0.61	15.08	86.10
England	1624	Tottenham Hotspur	38	1.95	0.95	1.00	15.79	84.95
England	1611	Manchester United	38	1.79	0.74	1.05	12.87	84.79
France	3767	PSG	38	2.84	0.76	2.08	15.89	89.16
France	3766	Olympique Lyonnais	38	2.29	1.13	1.16	14.92	86.25
France	19830	Monaco	38	2.24	1.18	1.05	12.42	82.52
France	3771	Olympique Marseille	38	2.11	1.24	0.87	15.53	85.31
France	3804	Dijon	38	1.45	1.92	-0.47	11.18	81.32
Germany	2444	Bayern München	34	2.71	0.82	1.88	16.62	87.83
Germany	2482	Hoffenheim	34	1.94	1.41	0.53	13.53	82.85
Germany	2447	Borussia Dortmund	34	1.88	1.38	0.50	13.15	84.31
Germany	2446	Bayer Leverkusen	34	1.71	1.29	0.41	14.12	84.15
Germany	2975	RB Leipzig	34	1.68	1.56	0.12	13.53	81.43
Italy	3162	Lazio	38	2.34	1.29	1.05	14.08	83.72
Italy	3159	Juventus	38	2.26	0.63	1.63	13.95	87.96
Italy	3187	Napoli	38	2.03	0.76	1.26	16.66	87.87
Italy	3161	Internazionale	38	1.74	0.79	0.95	14.68	85.58
Italy	3158	Roma	38	1.61	0.74	0.87	16.63	84.12
Spain	676	Barcelona	38	2.61	0.76	1.84	14.89	88.35
Spain	675	Real Madrid	38	2.47	1.16	1.32	17.61	87.92
Spain	687	Real Sociedad	38	1.74	1.55	0.18	12.50	85.79
Spain	674	Valencia	38	1.71	1.00	0.71	11.89	83.54
Spain	684	Real Betis	38	1.58	1.61	-0.03	10.08	86.19

```
ggplot(team_summary,
       aes(x = reorder(team_name, goals_per_match),
           y = goals_per_match, color = league)) +
  geom_point(size = 2.5, show.legend = FALSE) +
  facet_wrap(~ league, scales = "free_y") +
  coord_flip() +
  scale_color_manual(values = league_palette) +
  labs(
    title = "Team scoring averages within each league",
    x = NULL, y = "Goals per team-match"
  ) +
  theme_eda
```



Team averages are based on the available team-match records. This plot is descriptive and does not adjust for strength of schedule, injuries, managerial changes, or other contextual factors.

7 6. Home versus away performance

7.1 6.1 Supplied home/away summary

```
knitr::kable(
  home_away_summary,
  digits = 2,
  caption = "Provided league-by-venue summary"
)
```

Table 15: Provided league-by-venue summary

league	location	team_matches	avg_goals	avg_shots	avg_pass_accuracy	avg_pass_share
England	Away	380	1.15	10.36	80.65	48.28
England	Home	380	1.53	13.01	81.13	51.72
France	Away	380	1.19	10.67	81.82	48.56
France	Home	380	1.53	12.96	82.27	51.44
Germany	Away	306	1.19	10.57	80.73	48.92
Germany	Home	306	1.60	13.25	81.24	51.08
Italy	Away	380	1.22	11.04	82.00	49.06
Italy	Home	380	1.46	13.56	82.21	50.94
Spain	Away	380	1.15	9.84	81.26	48.58
Spain	Home	380	1.55	12.64	81.99	51.42

7.2 6.2 Recalculate home/away averages from the primary data

```
home_away_recalc <- team_match |>
  group_by(league, venue) |>
  summarise(
    team_matches = n(),
    avg_goals = mean(goals, na.rm = TRUE),
    avg_shots = mean(shots, na.rm = TRUE),
    avg_pass_accuracy = mean(pass_accuracy_pct, na.rm = TRUE),
    avg_pass_share = mean(pass_share_pct, na.rm = TRUE),
    .groups = "drop"
  )

knitr::kable(home_away_recalc, digits = 2,
             caption = "Home/away averages recalculated from team-match data")
```

Table 16: Home/away averages recalculated from team-match data

league	venue	team_matches	avg_goals	avg_shots	avg_pass_accuracy	avg_pass_share
England	Away	380	1.15	10.36	80.65	48.28
England	Home	380	1.53	13.01	81.13	51.72
France	Away	380	1.19	10.67	81.82	48.56
France	Home	380	1.53	12.96	82.27	51.44
Germany	Away	306	1.19	10.57	80.73	48.92
Germany	Home	306	1.60	13.25	81.24	51.08
Italy	Away	380	1.22	11.04	82.00	49.06
Italy	Home	380	1.46	13.56	82.21	50.94
Spain	Away	380	1.15	9.84	81.26	48.58
Spain	Home	380	1.55	12.64	81.99	51.42

Recalculating the summary from the primary dataset provides a reproducibility check. Small differences from the supplied summary may result from rounding or different handling of missing observations; large differences would require investigation.

7.3 6.3 Home-away differences by league

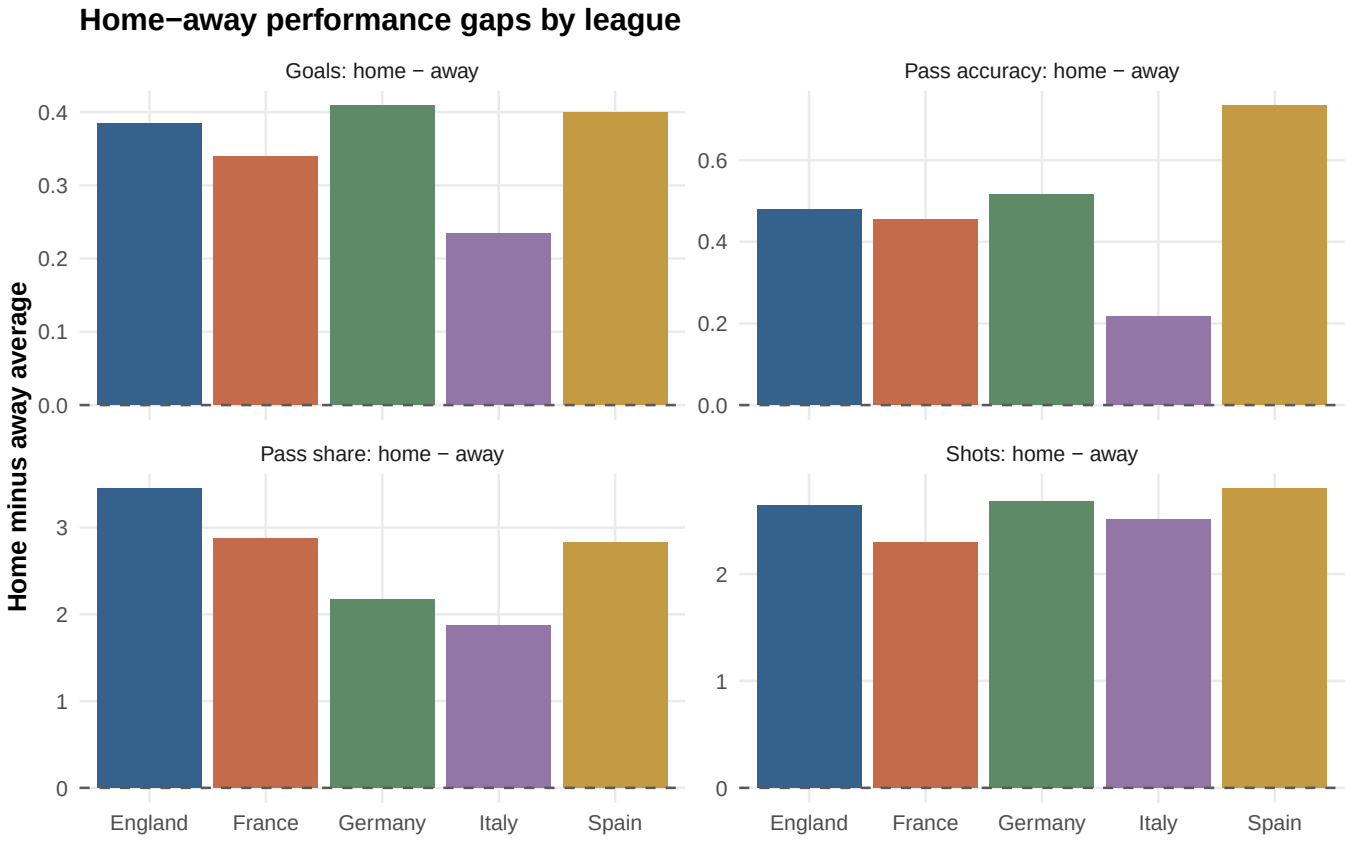
```
home_away_diff <- home_away_recalc |>
  pivot_longer(
    cols = starts_with("avg_"),
    names_to = "metric", values_to = "value"
  ) |>
  select(league, venue, metric, value) |>
  pivot_wider(names_from = venue, values_from = value) |>
  mutate(home_minus_away = Home - Away)

knitr::kable(home_away_diff, digits = 3,
             caption = "Home minus away differences by league")
```

Table 17: Home minus away differences by league

league	metric	Away	Home	home_minus_away
England	avg_goals	1.147	1.532	0.384
England	avg_shots	10.363	13.008	2.645
England	avg_pass_accuracy	80.649	81.127	0.478
England	avg_pass_share	48.276	51.724	3.449
France	avg_goals	1.189	1.529	0.339
France	avg_shots	10.666	12.958	2.292
France	avg_pass_accuracy	81.815	82.269	0.454
France	avg_pass_share	48.564	51.436	2.871
Germany	avg_goals	1.193	1.601	0.408
Germany	avg_shots	10.572	13.252	2.680
Germany	avg_pass_accuracy	80.727	81.243	0.516
Germany	avg_pass_share	48.916	51.084	2.168
Italy	avg_goals	1.221	1.455	0.234
Italy	avg_shots	11.042	13.555	2.513
Italy	avg_pass_accuracy	81.996	82.213	0.217
Italy	avg_pass_share	49.064	50.936	1.871
Spain	avg_goals	1.147	1.547	0.400
Spain	avg_shots	9.845	12.642	2.797
Spain	avg_pass_accuracy	81.258	81.991	0.733
Spain	avg_pass_share	48.584	51.416	2.831

```
ggplot(
  home_away_diff |> filter(metric %in% c("avg_goals", "avg_shots",
                                          "avg_pass_accuracy", "avg_pass_share")),
  aes(league, home_minus_away, fill = league)
) +
  geom_col(show.legend = FALSE) +
  facet_wrap(
    ~ metric, scales = "free_y",
    labeller = as_labeller(c(
      avg_goals = "Goals: home - away",
      avg_shots = "Shots: home - away",
      avg_pass_accuracy = "Pass accuracy: home - away",
      avg_pass_share = "Pass share: home - away"
    ))
  ) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey35") +
  scale_fill_manual(values = league_palette) +
  labs(
    title = "Home-away performance gaps by league",
    x = NULL, y = "Home minus away average"
  ) +
  theme_eda
```



A positive value indicates that the home side has a higher average for that metric. For pass share, the comparison is a share-of-passes proxy: the home side's higher share necessarily corresponds to a lower share for the away side in the same match.

7.4 6.4 Paired match-level home advantage

Because each match has both a home and an away record, a paired comparison is more informative than treating all home and away rows as unrelated. The next section constructs one row per match and calculates home-minus-away differences.

```
match_pairs <- team_match |>
  select(match_id, league, match_date, venue, goals, shots,
         passes_attempted, pass_accuracy_pct, pass_share_pct) |>
  pivot_wider(
    names_from = venue,
    values_from = c(goals, shots, passes_attempted,
                    pass_accuracy_pct, pass_share_pct)
  ) |>
  mutate(
    goal_gap = goals_Home - goals_Away,
    shot_gap = shots_Home - shots_Away,
    pass_gap = passes_attempted_Home - passes_attempted_Away,
    accuracy_gap = pass_accuracy_pct_Home - pass_accuracy_pct_Away,
    pass_share_gap = pass_share_pct_Home - pass_share_pct_Away
  )

paired_summary <- match_pairs |>
  summarise(
    matches = n(),
```

```
    home_wins = sum(goal_gap > 0, na.rm = TRUE),
    draws = sum(goal_gap == 0, na.rm = TRUE),
    away_wins = sum(goal_gap < 0, na.rm = TRUE),
    home_win_pct = 100 * home_wins / matches,
    draw_pct = 100 * draws / matches,
    away_win_pct = 100 * away_wins / matches,
    mean_goal_gap = mean(goal_gap, na.rm = TRUE),
    median_goal_gap = median(goal_gap, na.rm = TRUE)
  )

knitr::kable(paired_summary, digits = 2,
             caption = "Paired home-versus-away match outcomes")
```

Table 18: Paired home-versus-away match outcomes

matches	home_wins	draws	away_wins	home_win_pct	draw_pct	away_win_pct	mean_goal_gap	median_goal_gap
1826	828	447	551	45.35	24.48	30.18	0.35	0

```
match_pairs |>
  summarise(
    across(
      c(goal_gap, shot_gap, pass_gap, accuracy_gap, pass_share_gap),
      list(mean = ~ mean(.x, na.rm = TRUE),
           median = ~ median(.x, na.rm = TRUE),
           sd = ~ sd(.x, na.rm = TRUE))
    )
  ) |>
  pivot_longer(everything(), names_to = "statistic", values_to = "value") |>
  knitr::kable(digits = 2, caption = "Distribution summaries of paired match gaps")
```

Table 19: Distribution summaries of paired match gaps

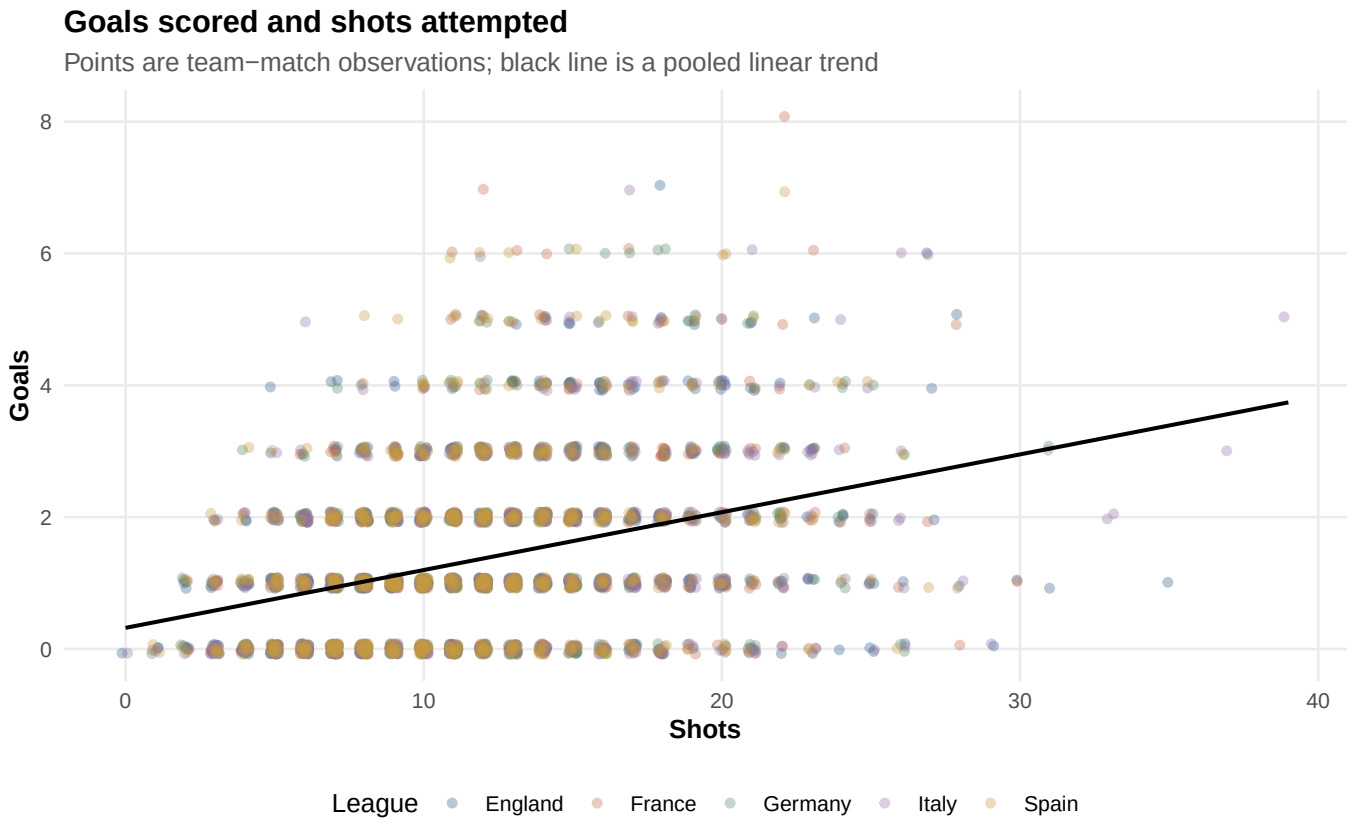
statistic	value
goal_gap_mean	0.35
goal_gap_median	0.00
goal_gap_sd	1.86
shot_gap_mean	2.58
shot_gap_median	2.50
shot_gap_sd	7.89
pass_gap_mean	24.10
pass_gap_median	22.50
pass_gap_sd	223.93
accuracy_gap_mean	0.48
accuracy_gap_median	0.63
accuracy_gap_sd	7.89
pass_share_gap_mean	2.66
pass_share_gap_median	2.95
pass_share_gap_sd	24.81

The paired design controls for the fact that both teams experienced the same match. It does not, by itself, control for team quality or the selection of which teams played at home in a particular fixture.

8 7. Relationships among performance measures

8.1 7.1 Goals and shots

```
ggplot(team_match, aes(shots, goals, color = league)) +
  geom_jitter(alpha = 0.35, width = 0.15, height = 0.08) +
  geom_smooth(method = "lm", se = FALSE, color = "black",
              linewidth = 0.8) +
  scale_color_manual(values = league_palette) +
  labs(
    title = "Goals scored and shots attempted",
    subtitle = "Points are team-match observations; black line is a pooled linear trend",
    x = "Shots", y = "Goals", color = "League"
  ) +
  theme_eda
```




```
goals_shots_cor <- cor.test(
  team_match$shots, team_match$goals,
  method = "spearman", exact = FALSE
)

tibble(
  method = "Spearman rank correlation",
  estimate = unname(goals_shots_cor$estimate),
  p_value = goals_shots_cor$p.value
) |>
  knitr::kable(digits = 4, caption = "Association between shots and goals")
```

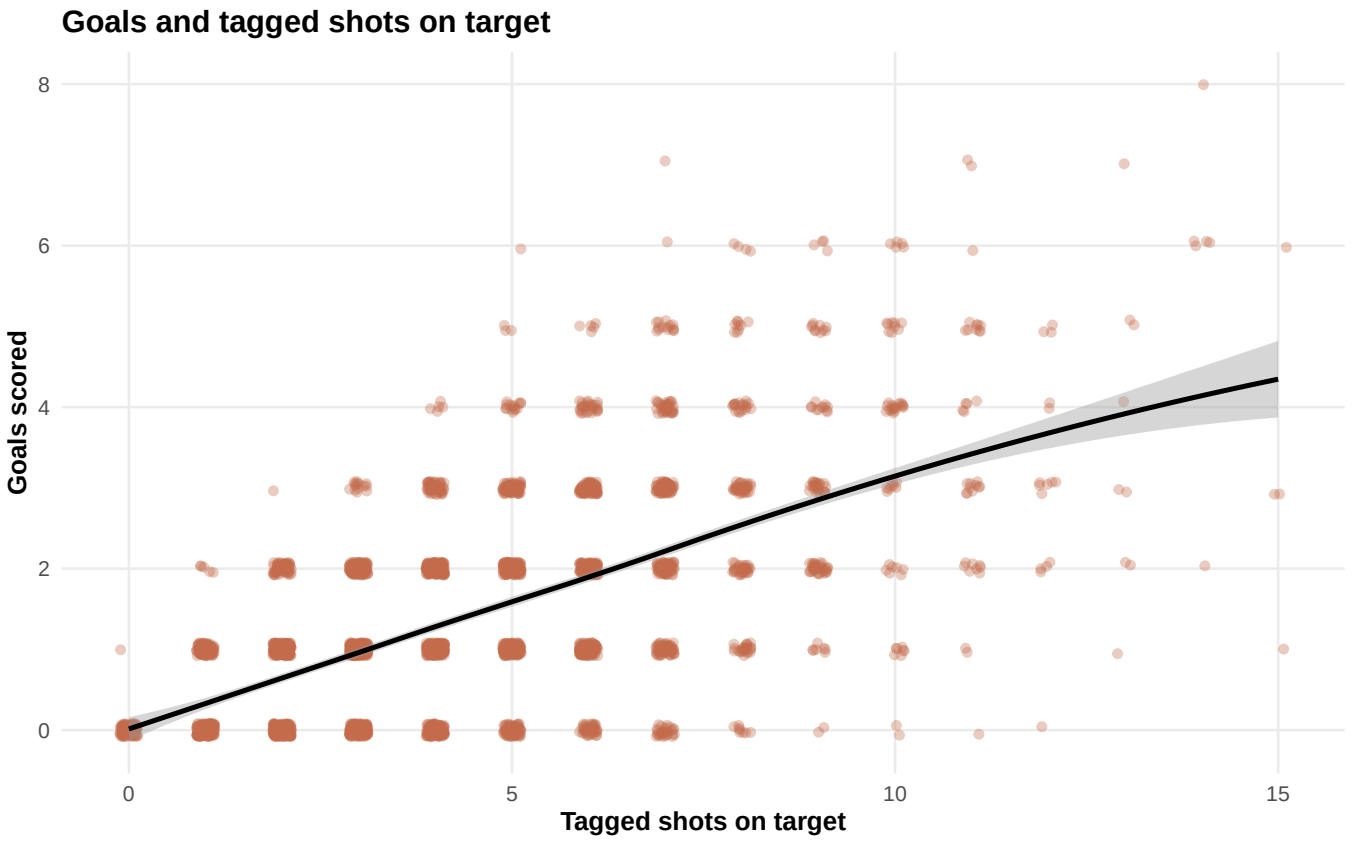
Table 20: Association between shots and goals

method	estimate	p_value
Spearman rank correlation	0.3434	0

A positive association is expected: teams that generate more shots generally have more opportunities to score. However, shot quality is not represented in the supplied team-match table. A team can take many low-probability shots while another creates fewer but better chances.

8.2 7.2 Goals and tagged shots on target

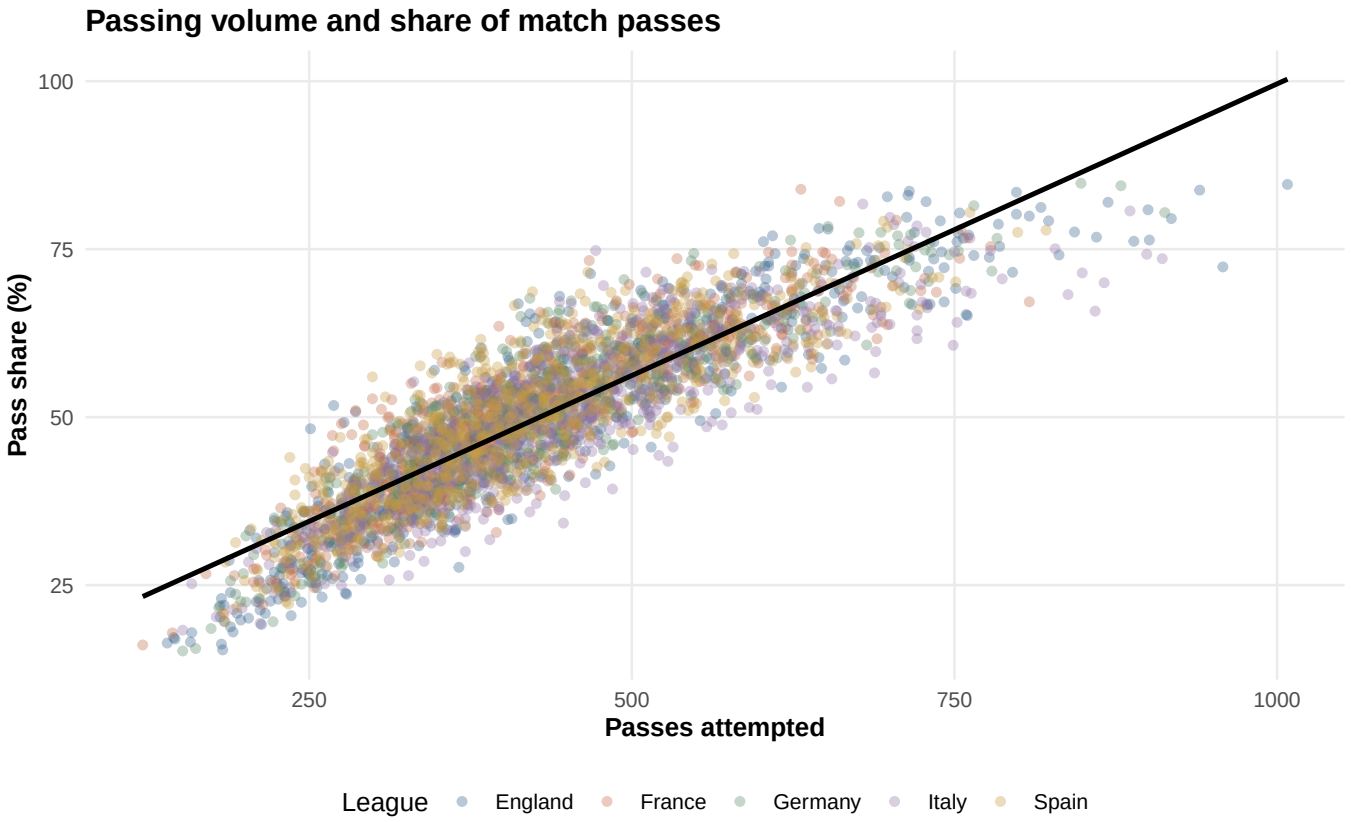
```
ggplot(team_match, aes(on_target_tagged, goals)) +
  geom_jitter(alpha = 0.35, color = "#C36B4B",
              width = 0.12, height = 0.08) +
  geom_smooth(method = "loess", se = TRUE, color = "black") +
  labs(
    title = "Goals and tagged shots on target",
    x = "Tagged shots on target", y = "Goals scored"
  ) +
  theme_eda
```



This relationship is especially sensitive to the definition and completeness of the event tag. The shot-type audit later in the report helps assess whether goals and tagged accuracy align across different shot categories.

8.3 7.3 Passing volume and pass share

```
ggplot(team_match, aes(passes_attempted, pass_share_pct, color = league)) +
  geom_point(alpha = 0.35) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  scale_color_manual(values = league_palette) +
  labs(
    title = "Passing volume and share of match passes",
    x = "Passes attempted", y = "Pass share (%)",
    color = "League"
  ) +
  theme_eda
```

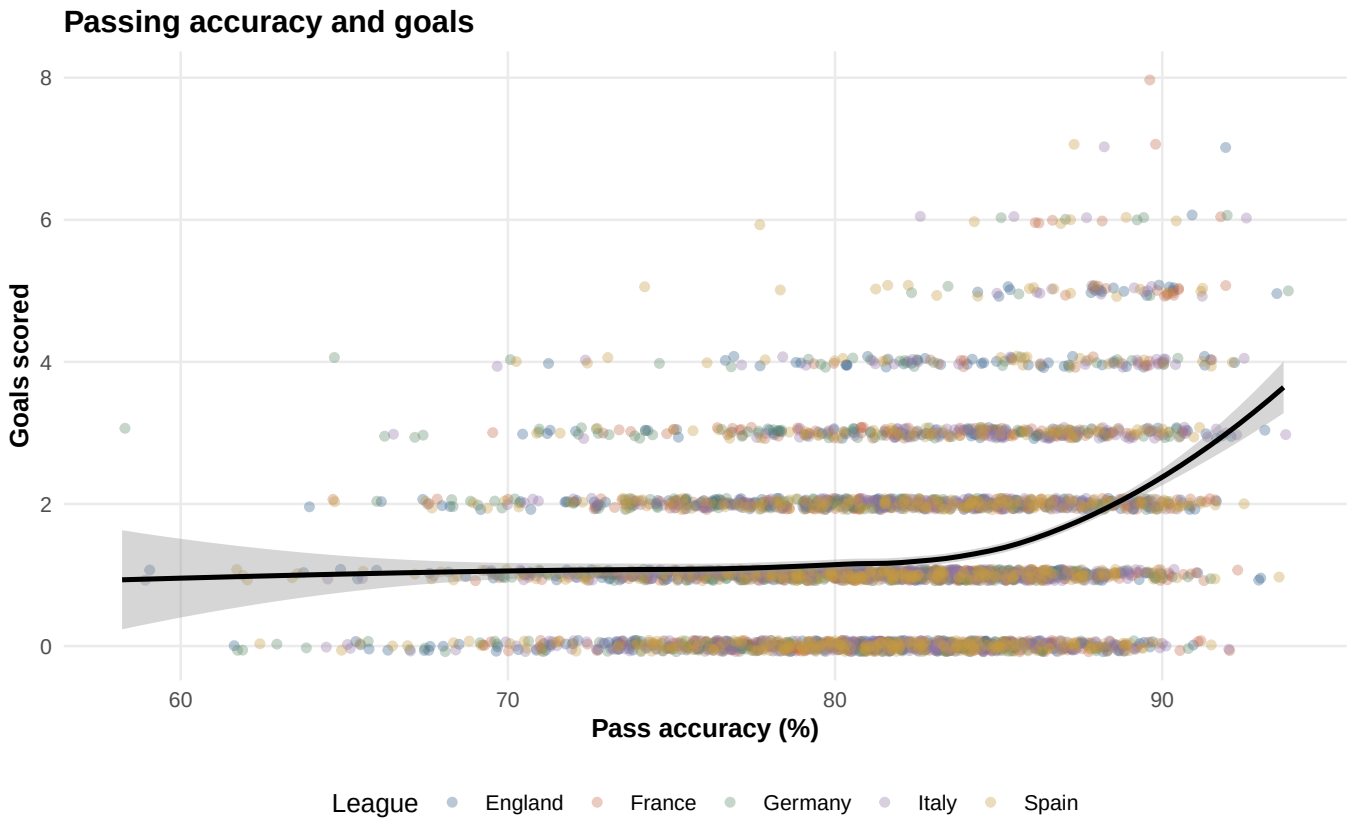


Pass share is mathematically related to passing volume because it is calculated using the team's passes relative to both teams' passes. A strong relationship between these two variables is therefore partly mechanical, not necessarily an independent tactical discovery.

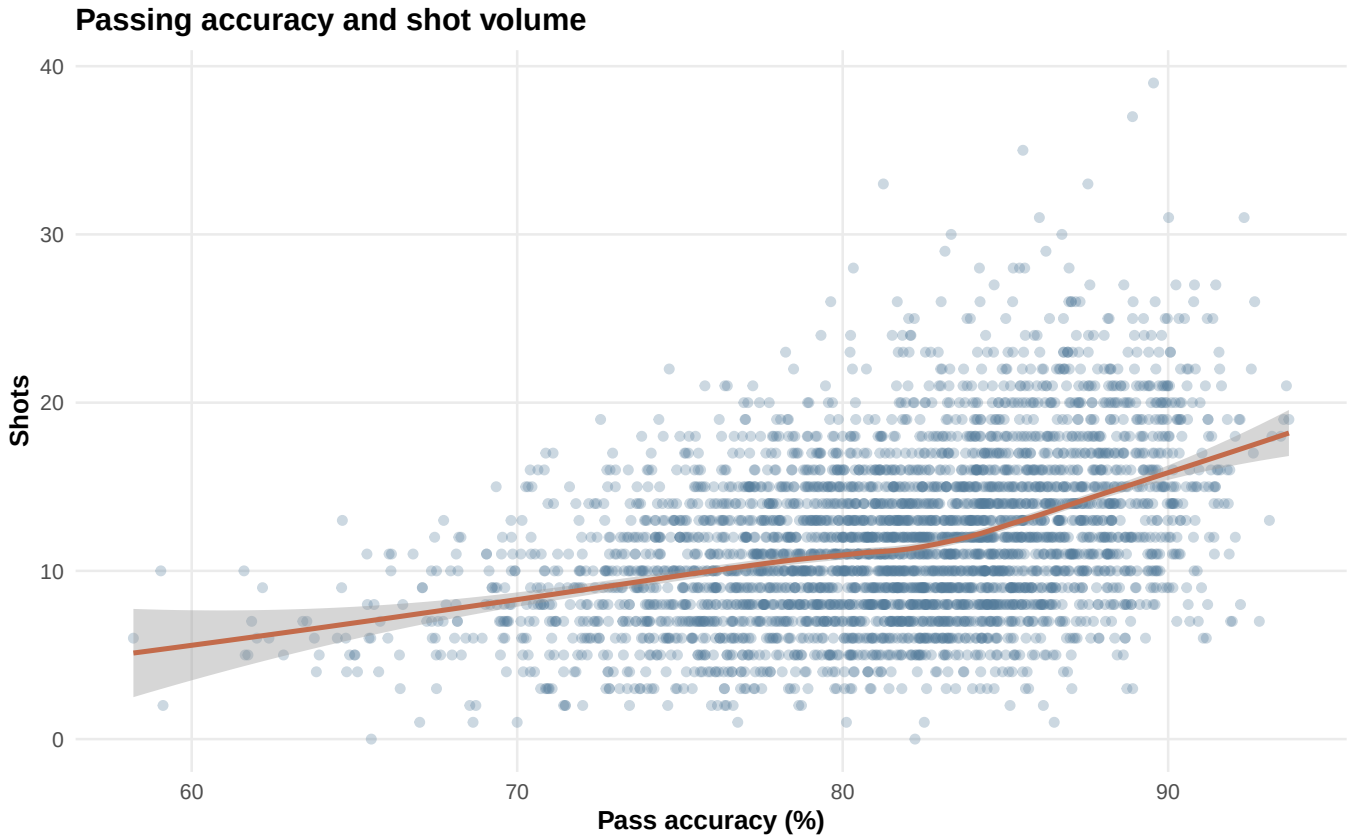
8.4 7.4 Passing accuracy and attacking output

```
ggplot(team_match, aes(pass_accuracy_pct, goals, color = league)) +
  geom_jitter(alpha = 0.35, width = 0.15, height = 0.08) +
  geom_smooth(method = "loess", se = TRUE, color = "black") +
  scale_color_manual(values = league_palette) +
  labs(
    title = "Passing accuracy and goals",
    x = "Pass accuracy (%)", y = "Goals scored",
    color = "League"
```

```
  ) +
  theme_eda
```




```
ggplot(team_match, aes(pass_accuracy_pct, shots)) +
  geom_point(alpha = 0.3, color = "#547C9B") +
  geom_smooth(method = "loess", se = TRUE, color = "#C36B4B") +
  labs(
    title = "Passing accuracy and shot volume",
    x = "Pass accuracy (%)", y = "Shots"
  ) +
  theme_eda
```



Passing accuracy should not automatically be interpreted as attacking quality. Teams may achieve high completion rates through safer, shorter passes, while teams attempting riskier forward passes may have lower accuracy but create more chances.

8.5 7.5 Correlation matrix

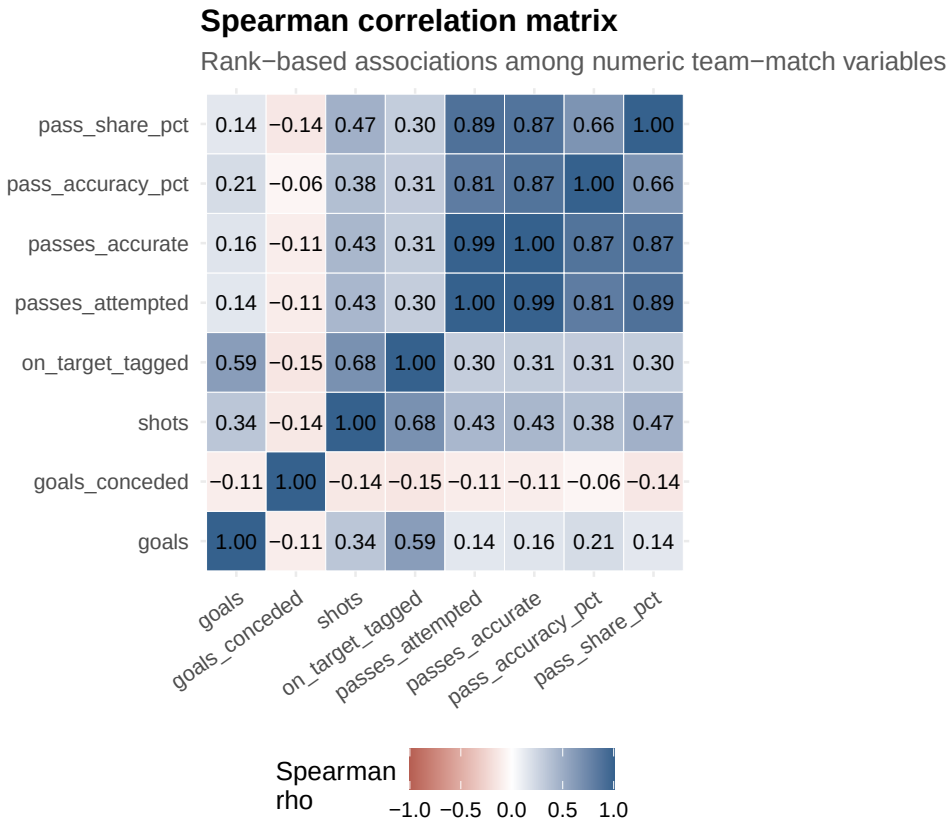
```
cor_data <- team_match |>
  select(all_of(numeric_vars)) |>
  drop_na()

cor_matrix <- cor(cor_data, method = "spearman")

cor_long <- as.data.frame(as.table(cor_matrix)) |>
  rename(variable_x = Var1, variable_y = Var2, rho = Freq)

ggplot(cor_long, aes(variable_x, variable_y, fill = rho)) +
  geom_tile(color = "white") +
  geom_text(aes(label = sprintf("%.2f", rho)), size = 3) +
  scale_fill_gradient2(
    low = "#B65E52", mid = "white", high = "#35618D",
    midpoint = 0, limits = c(-1, 1)
  ) +
  coord_equal() +
  labs(
    title = "Spearman correlation matrix",
    subtitle = "Rank-based associations among numeric team-match variables",
    x = NULL, y = NULL, fill = "Spearman\nrho"
  ) +
```

```
  theme_eda +
  theme(axis.text.x = element_text(angle = 35, hjust = 1))
```




```
cor_pairs <- cor_long |>
  filter(as.character(variable_x) < as.character(variable_y)) |>
  arrange(desc(abs(rho)))

knitr::kable(cor_pairs, digits = 3,
             caption = "Unique pairwise Spearman correlations")
```

Table 21: Unique pairwise Spearman correlations

variable_x	variable_y	rho
passes_accurate	passes_attempted	0.994
pass_share_pct	passes_attempted	0.887
pass_accuracy_pct	passes_accurate	0.871
pass_share_pct	passes_accurate	0.868
pass_accuracy_pct	passes_attempted	0.812
on_target_tagged	shots	0.678
pass_accuracy_pct	pass_share_pct	0.657
goals	on_target_tagged	0.595
pass_share_pct	shots	0.475
passes_accurate	shots	0.434
passes_attempted	shots	0.430
pass_accuracy_pct	shots	0.385
goals	shots	0.343
on_target_tagged	pass_accuracy_pct	0.308

variable_x	variable_y	rho
on_target_tagged	passes_accurate	0.306
on_target_tagged	pass_share_pct	0.300
on_target_tagged	passes_attempted	0.296
goals	pass_accuracy_pct	0.209
goals	passes_accurate	0.157
goals_conceded	on_target_tagged	-0.153
goals	passes_attempted	0.142
goals	pass_share_pct	0.142
goals_conceded	pass_share_pct	-0.142
goals_conceded	shots	-0.138
goals_conceded	passes_attempted	-0.114
goals	goals_conceded	-0.109
goals_conceded	passes_accurate	-0.106
goals_conceded	pass_accuracy_pct	-0.060

Correlation is a measure of association, not causation. The correlation matrix also includes mathematically linked variables. For example, accurate passes contribute to passing accuracy, and team pass volume contributes to pass share. Those relationships should be interpreted with care.

9 8. Outliers and unusual observations

Outliers are not necessarily errors. In football, a high-scoring match, unusually large shot count, or very high passing total may represent a genuine performance. The purpose of this section is to identify unusual values for review, not to delete them automatically.

9.1 8.1 IQR-based outlier counts

```
iqr_outliers <- map_dfr(numeric_vars, function(v) {
  x <- team_match[[v]]
  q1 <- quantile(x, 0.25, na.rm = TRUE)
  q3 <- quantile(x, 0.75, na.rm = TRUE)
  iqr <- q3 - q1
  lower <- q1 - 1.5 * iqr
  upper <- q3 + 1.5 * iqr

  tibble(
    variable = v,
    q1 = q1,
    q3 = q3,
    iqr = iqr,
    lower_fence = lower,
    upper_fence = upper,
    outlier_count = sum(x < lower | x > upper, na.rm = TRUE),
    outlier_pct = 100 * mean(x < lower | x > upper, na.rm = TRUE)
  )
})

knitr::kable(iqr_outliers, digits = 2,
             caption = "IQR rule: potential univariate outliers")
```

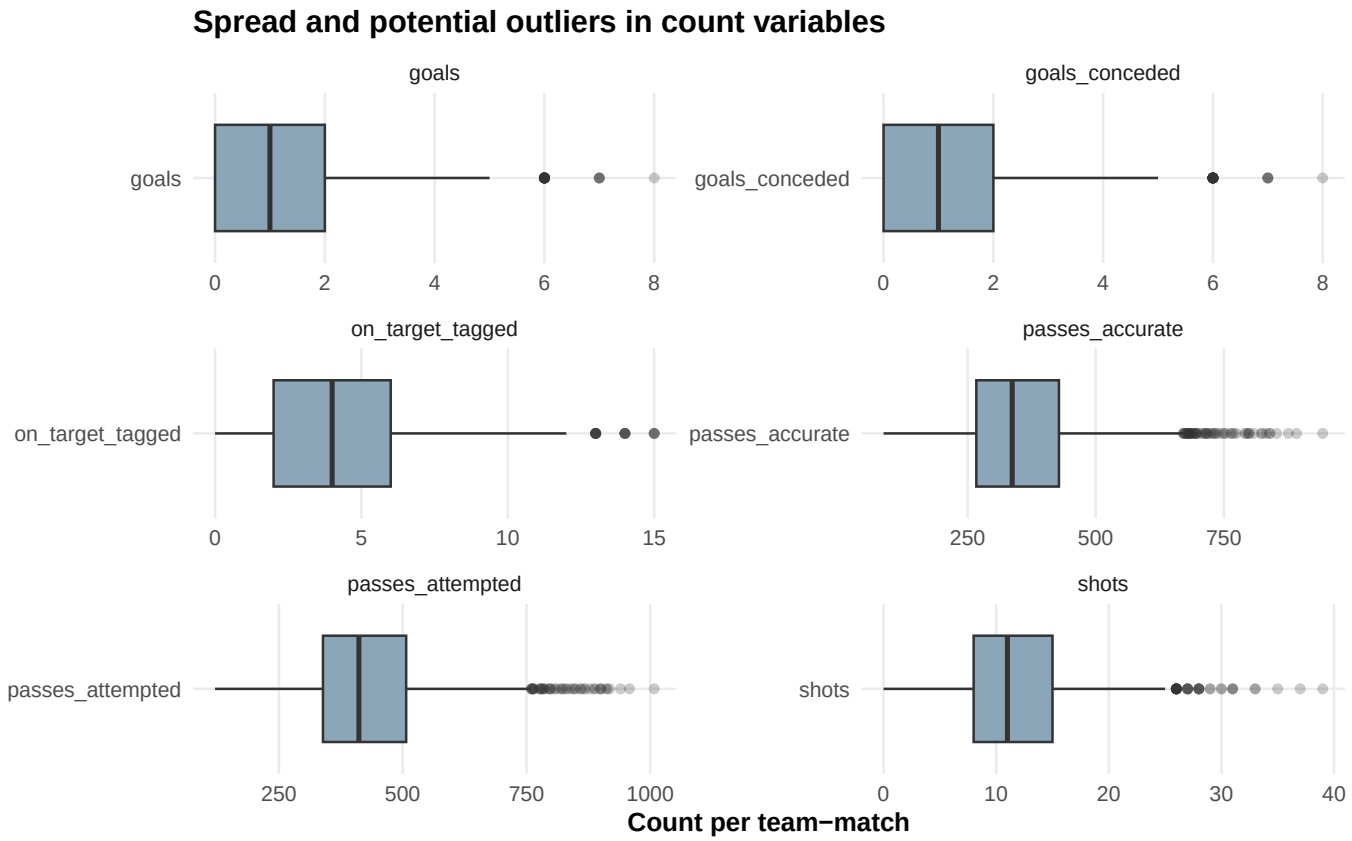
Table 22: IQR rule: potential univariate outliers

variable	q1	q3	iqr	lower_fence	upper_fence	outlier_count	outlier_pct
goals	0.00	2.00	2.00	-3.00	5.00	26	0.71
goals_conceded	0.00	2.00	2.00	-3.00	5.00	26	0.71
shots	8.00	15.00	7.00	-2.50	25.50	38	1.04
on_target_tagged	2.00	6.00	4.00	-4.00	12.00	19	0.52
passes_attempted	339.00	507.00	168.00	87.00	759.00	47	1.29
passes_accurate	267.00	428.25	161.25	25.12	670.12	59	1.62
pass_accuracy_pct	78.18	85.46	7.28	67.26	96.38	45	1.23
pass_share_pct	40.97	59.03	18.05	13.90	86.10	0	0.00

9.2 8.2 Boxplots for count measures

```
count_long <- team_match |>
  select(goals, goals_conceded, shots, on_target_tagged,
         passes_attempted, passes_accurate) |>
  pivot_longer(everything(), names_to = "variable", values_to = "value")

ggplot(count_long, aes(variable, value)) +
  geom_boxplot(fill = "#8BA6B8", outlier.alpha = 0.25) +
  coord_flip() +
  facet_wrap(~ variable, scales = "free", ncol = 2) +
  labs(
    title = "Spread and potential outliers in count variables",
    x = NULL, y = "Count per team-match"
  ) +
  theme_eda
```



9.3 8.3 Highest observed values

```
extreme_team_matches <- team_match |>
  arrange(desc(goals), desc(shots)) |>
  select(match_date, league, team_name, opponent_name, venue,
         goals, goals_conceded, shots, on_target_tagged,
         passes_attempted, pass_accuracy_pct) |>
  slice_head(n = 15)

knitr::kable(extreme_team_matches, digits = 2,
             caption = "Team-match observations with the highest goal totals")
```

Table 23: Team-match observations with the highest goal totals

match_date	league	team_name	opponent_name	venue	goals	goals_conceded	shots	on_target_tagged	passes_attempted	pass_accuracy_pct
2018-01-17	France	PSG	Dijon	Home	8	0	22	14	636	89.62
2018-01-21	Spain	Real Madrid	Deportivo La Coruña	Home	7	1	22	13	583	87.31
2017-10-14	England	Manchester City	Stoke City	Home	7	2	18	11	869	92.06
2018-02-04	Italy	Juventus	Sassuolo	Home	7	0	17	11	660	88.18
2018-04-15	France	PSG	Monaco	Home	7	1	12	7	516	89.92

match_date	league	team_name	opponent_name	venue	goals	goals_conceded	shots	on_target_tagged	passes_attempted	pass_accuracy_pct
2017-09-16	England	Manchester City	Watford	Away	6	0	27	10	631	90.81
2018-03-31	Italy	Lazio	Benevento	Home	6	2	27	15	524	87.60
2017-09-17	Italy	Napoli	Benevento	Home	6	0	26	14	899	92.66
2017-08-20	France	PSG	Toulouse	Home	6	2	23	14	624	86.06
2017-10-01	Italy	Lazio	Sassuolo	Home	6	1	21	10	530	85.47
2018-02-24	Spain	Barcelona	Girona	Home	6	1	20	14	581	87.26
2018-03-18	Spain	Real Madrid	Girona	Home	6	3	20	14	612	90.36
2018-03-10	Germany	Bayern München	Hamburger SV	Home	6	0	18	8	676	89.20
2018-03-31	Germany	Hoffenheim	Köln	Home	6	0	18	11	368	85.05
2017-11-04	France	Monaco	Guingamp	Home	6	0	17	9	421	86.22

Extreme observations should be checked against original match or event records if they materially affect later conclusions. Do not remove them solely because they exceed an IQR fence.

10 9. Shot-type audit and measurement validity

The shot-type audit separates free-kick shots, open-play shots, and penalties. This matters because a penalty is a different type of scoring opportunity from an open-play attempt, and event tags may be assigned differently across event types.

10.1 9.1 Audit table

```
shot_audit |>
  mutate(
    tagged_accuracy_rate = 100 * tagged_accurate / attempts,
    goal_per_attempt_rate = 100 * tagged_goals / attempts,
    goals_without_tag_rate = 100 * goals_without_accurate_tag / attempts
  ) |>
  knitr::kable(digits = 2,
               caption = "Shot-event audit with derived rates")
```

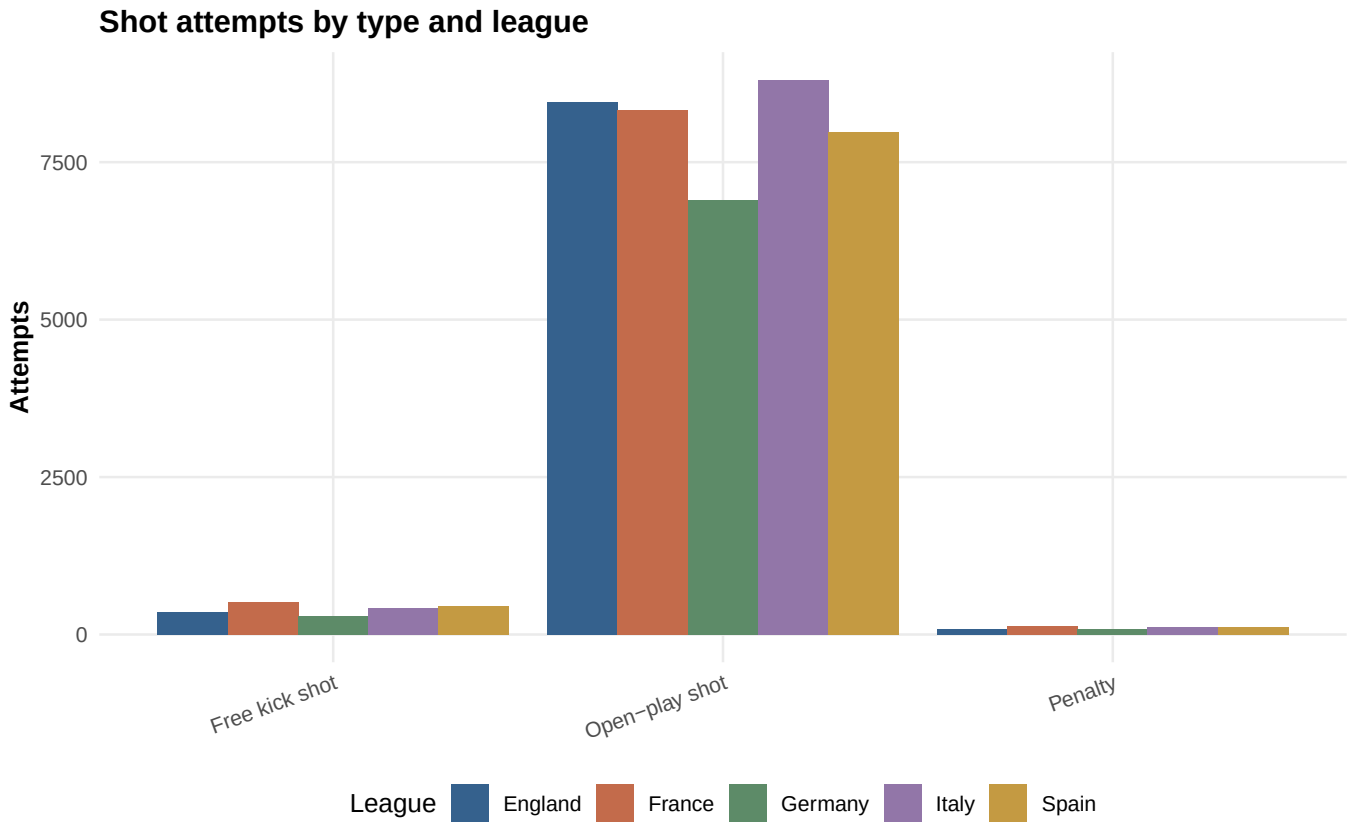
Table 24: Shot-event audit with derived rates

league	shot_type	attempts	tagged_accurate	tagged_goals	goals_without_accurate_tag	tagged_accuracy_rate	goal_per_attempt_rate	goals_without_tag_rate
England	Free kick shot	350	100	18	0	28.57	5.14	0.00

league	shot_type	attempts	tagged_accurate	tagged_goals	goals_without_accurate_tag	tagged_accuracy_rate	goal_per_attempt_rate	goals_without_tag_rate
England	Open-play shot	8451	2986	914	2	35.33	10.82	0.02
England	Penalty	80	56	56	0	70.00	70.00	0.00
France	Free kick shot	521	163	26	0	31.29	4.99	0.00
France	Open-play shot	8327	2963	873	1	35.58	10.48	0.01
France	Penalty	129	99	99	0	76.74	76.74	0.00
Germany	Free kick shot	299	92	18	0	30.77	6.02	0.00
Germany	Open-play shot	6898	2512	747	1	36.42	10.83	0.01
Germany	Penalty	93	68	68	0	73.12	73.12	0.00
Italy	Free kick shot	415	143	36	0	34.46	8.67	0.00
Italy	Open-play shot	8806	3069	853	0	34.85	9.69	0.00
Italy	Penalty	126	89	89	0	70.63	70.63	0.00
Spain	Free kick shot	453	152	28	0	33.55	6.18	0.00
Spain	Open-play shot	7979	2960	884	0	37.10	11.08	0.00
Spain	Penalty	113	81	81	0	71.68	71.68	0.00

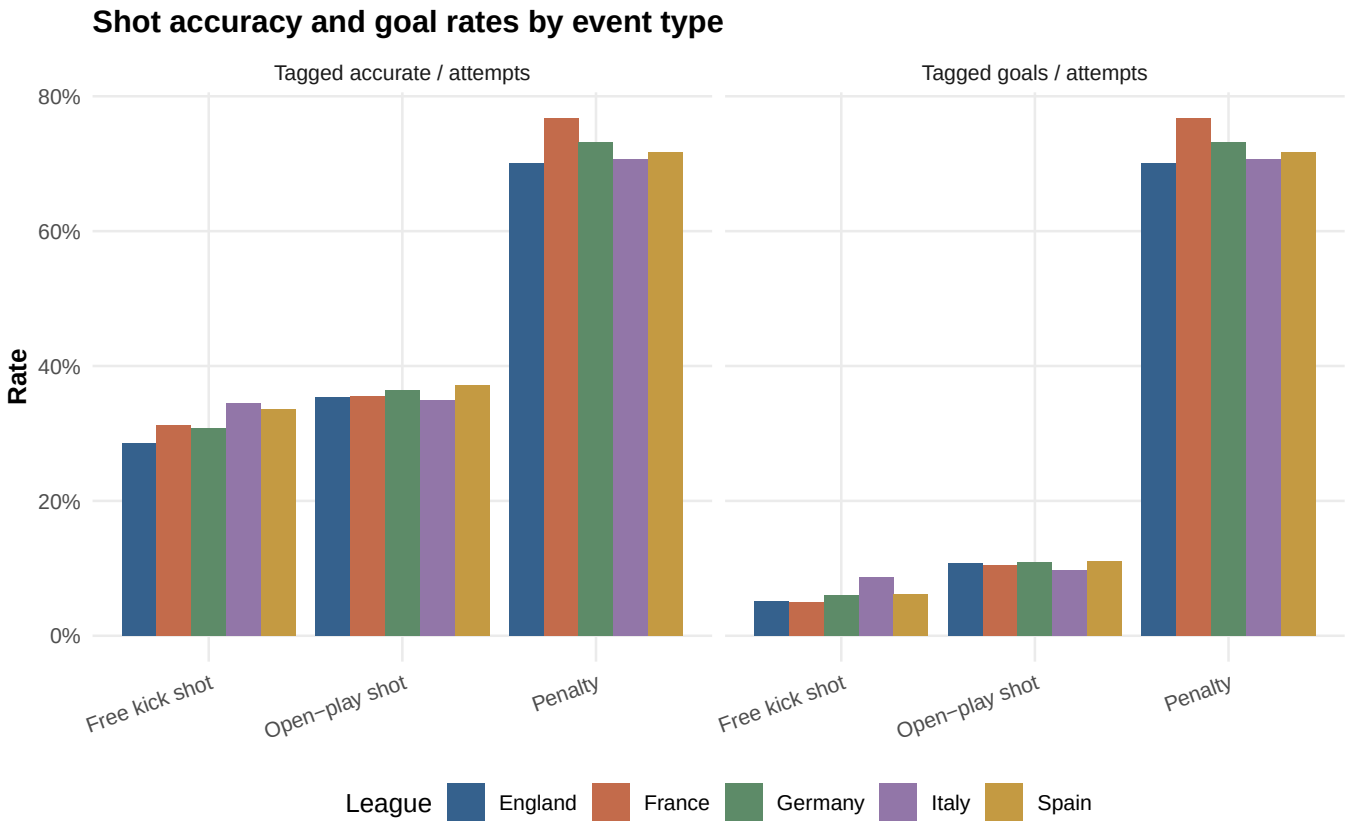
10.2 9.2 Attempts and tagged accuracy by shot type

```
ggplot(shot_audit, aes(shot_type, attempts, fill = league)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = league_palette) +
  labs(
    title = "Shot attempts by type and league",
    x = NULL, y = "Attempts", fill = "League"
  ) +
  theme_eda +
  theme(axis.text.x = element_text(angle = 20, hjust = 1))
```




```
shot_audit_rates <- shot_audit |>
  mutate(
    tagged_accuracy_rate = tagged_accurate / attempts,
    tagged_goal_rate = tagged_goals / attempts
  ) |>
  select(league, shot_type, tagged_accuracy_rate, tagged_goal_rate) |>
  pivot_longer(
    cols = c(tagged_accuracy_rate, tagged_goal_rate),
    names_to = "rate", values_to = "proportion"
  ) |>
  mutate(
    rate = recode(
      rate,
      tagged_accuracy_rate = "Tagged accurate / attempts",
      tagged_goal_rate = "Tagged goals / attempts"
    )
  )

ggplot(shot_audit_rates,
       aes(shot_type, proportion, fill = league)) +
  geom_col(position = "dodge") +
  facet_wrap(~ rate) +
  scale_y_continuous(labels = label_percent()) +
  scale_fill_manual(values = league_palette) +
  labs(
    title = "Shot accuracy and goal rates by event type",
    x = NULL, y = "Rate", fill = "League"
  ) +
  theme_eda +
  theme(axis.text.x = element_text(angle = 20, hjust = 1))
```



The audit shows that penalties are a distinct category and that the tagged-goal counts for penalties align with tagged-accurate counts in the supplied audit. Open-play shots account for most attempts and most goals. A small number of open-play goals appear without an accurate tag in some leagues, which is a reason to treat the accuracy tag as a proxy rather than a flawless outcome definition.

10.3 9.3 Goals without an accurate tag

```
shot_audit |>
  group_by(shot_type) |>
  summarise(
    attempts = sum(attempts),
    tagged_accurate = sum(tagged_accurate),
    tagged_goals = sum(tagged_goals),
    goals_without_accurate_tag = sum(goals_without_accurate_tag),
    .groups = "drop"
  ) |>
  knitr::kable(caption = "Pooled shot-type audit across leagues")
```

Table 25: Pooled shot-type audit across leagues

shot_type	attempts	tagged_accurate	tagged_goals	goals_without_accurate_tag
Free kick shot	2038	650	126	0
Open-play shot	40461	14490	4271	4
Penalty	541	393	393	0

This audit is an important data-definition check. Before using `on_target_tagged` as a formal response variable or as an input to a model, verify the event-tag logic against the original event data and the provider's event definitions.

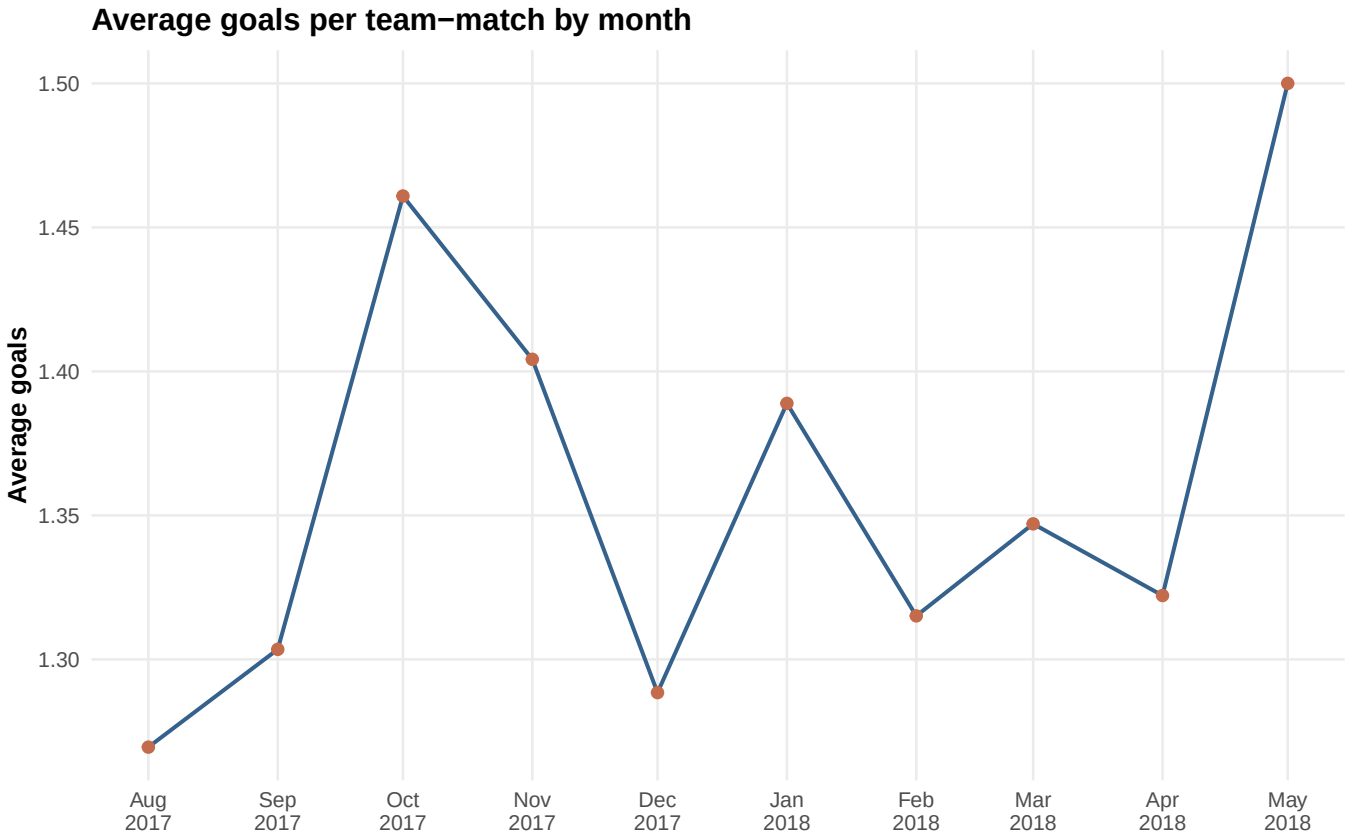
11 10. Time-oriented exploration

The dataset includes match dates, so it is possible to examine whether the available records show seasonal patterns. These summaries are descriptive and may reflect the competition calendar, postponements, and uneven match coverage.

11.1 10.1 Goals by month

```
monthly_goals <- team_match |>
  mutate(month = floor_date(match_date, "month")) |>
  group_by(month) |>
  summarise(
    team_matches = n(),
    goals_per_team_match = mean(goals, na.rm = TRUE),
    total_goals = sum(goals, na.rm = TRUE),
    .groups = "drop"
  )

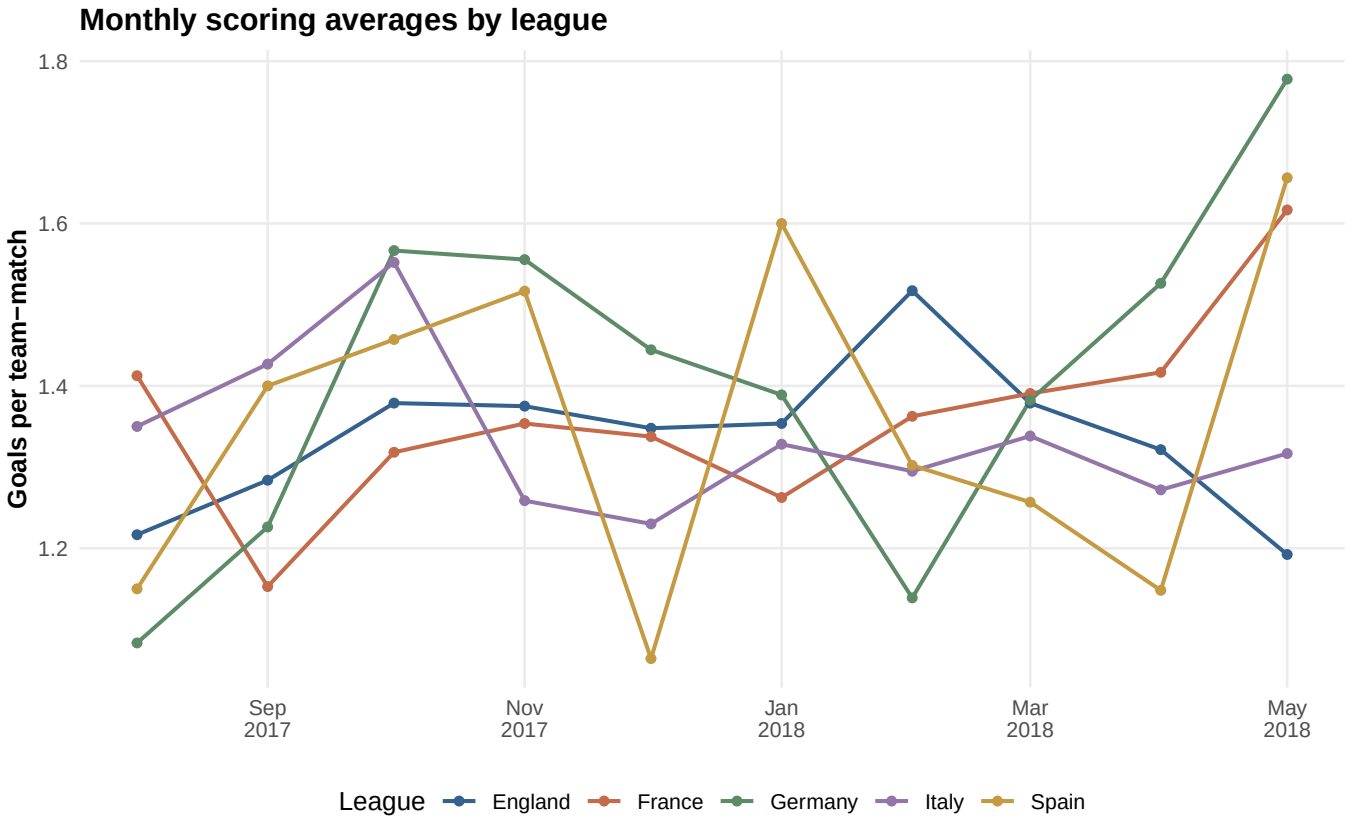
ggplot(monthly_goals, aes(month, goals_per_team_match)) +
  geom_line(linewidth = 0.8, color = "#35618D") +
  geom_point(size = 2, color = "#C36B4B") +
  scale_x_date(date_breaks = "1 month", date_labels = "%b\n%Y") +
  labs(
    title = "Average goals per team-match by month",
    x = NULL, y = "Average goals"
  ) +
  theme_eda
```



11.2 10.2 Monthly averages by league

```
monthly_league <- team_match |>
  mutate(month = floor_date(match_date, "month")) |>
  group_by(league, month) |>
  summarise(
    goals_per_team_match = mean(goals, na.rm = TRUE),
    shots_per_team_match = mean(shots, na.rm = TRUE),
    .groups = "drop"
  )

ggplot(monthly_league, aes(month, goals_per_team_match, color = league)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.5) +
  scale_color_manual(values = league_palette) +
  scale_x_date(date_breaks = "2 months", date_labels = "%b\n%Y") +
  labs(
    title = "Monthly scoring averages by league",
    x = NULL, y = "Goals per team-match", color = "League"
  ) +
  theme_eda
```



Month-level averages can be volatile when a league has few matches in a given month. Consider plotting the number of matches alongside the average before interpreting a spike or decline as a meaningful performance trend.

12 11. Interpretation and key findings

This section provides a structured synthesis of the descriptive patterns. The numerical summaries are calculated from the supplied team-match dataset and should be interpreted within the measurement limits already described.

```
overall_key_metrics <- team_match |>
  summarise(
    team_matches = n(),
    matches = n_distinct(match_id),
    goals_mean = mean(goals, na.rm = TRUE),
    goals_median = median(goals, na.rm = TRUE),
    shots_mean = mean(shots, na.rm = TRUE),
    on_target_mean = mean(on_target_tagged, na.rm = TRUE),
    pass_accuracy_mean = mean(pass_accuracy_pct, na.rm = TRUE),
    pass_share_mean = mean(pass_share_pct, na.rm = TRUE)
  )

league_goal_rank <- league_summary |>
  arrange(desc(avg_goals)) |>
  select(league, avg_goals, avg_shots, avg_pass_accuracy)

home_goal_gap <- home_away_diff |>
  filter(metric == "avg_goals") |>
  select(league, Away, Home, home_minus_away)
```

```
knitr::kable(overall_key_metrics, digits = 2,
             caption = "Key overall indicators")
```

Table 26: Key overall indicators

team_matches	matches	goals_mean	goals_median	shots_mean	on_target_mean	pass_accuracy_mean	pass_share_mean
3652	1826	1.35	1	11.79	4.25	81.55	50

```
knitr::kable(league_goal_rank, digits = 2,
             caption = "League ranking by average goals per team-match")
```

Table 27: League ranking by average goals per team-match

league	avg_goals	avg_shots	avg_pass_accuracy
Germany	1.40	11.91	80.98
France	1.36	11.81	82.04
Spain	1.35	11.24	81.62
England	1.34	11.69	80.89
Italy	1.34	12.30	82.10

```
knitr::kable(home_goal_gap, digits = 3,
             caption = "Home-away scoring comparison")
```

Table 28: Home-away scoring comparison

league	Away	Home	home_minus_away
England	1.147	1.532	0.384
France	1.189	1.529	0.339
Germany	1.193	1.601	0.408
Italy	1.221	1.455	0.234
Spain	1.147	1.547	0.400

12.0.1 Main descriptive takeaways

- **Scoring is a low-count outcome.** The average is higher than the median, indicating that a smaller number of high-scoring team performances influence the mean.
- **Shot volume varies considerably.** The distribution and upper tail show that some teams generate substantially more attempts than a typical team-match.
- **Passing accuracy and passing volume describe different aspects of play.** A high completion percentage does not necessarily mean a team attempted many passes or created more chances.
- **Home and away comparisons should be paired by match.** The paired match analysis uses the fact that every match contains both a home and away team record.
- **League averages summarize different competition environments.** They should not be interpreted as isolated causal effects of league membership.
- **The pass-share variable is a proxy.** Its average of 50% is an arithmetic consequence of the two teams' shares summing to 100% in each match.
- **The event-tag audit matters.** `on_target_tagged` is not automatically interchangeable with a provider's official shots-on-target statistic.

13 12. Limitations and cautions

1. **Single season:** The dataset represents 2017–18 only. It cannot establish whether the observed patterns persist across seasons.
2. **Team-match dependence:** Two rows belong to each match. Standard statistical tests that assume every row is independent may understate uncertainty.
3. **No player-level context:** The supplied analytical table does not describe lineups, substitutions, injuries, or individual player contributions.
4. **No shot-quality information:** Shot location, body part, defensive pressure, and expected goals are not included in the main team-match file.
5. **Event-tag definitions:** `on_target_tagged` is based on an event tag. Its validity should be checked against the underlying event definitions.
6. **Passing data are event-based:** Pass attempts exclude set-piece categories according to the data dictionary. This may not match every external passing statistic.
7. **Pass share is not possession time:** It measures a team's share of recorded passes, not the percentage of time it controlled the ball.
8. **League sample sizes differ:** Germany has fewer matches because its league has fewer teams in the season represented.
9. **Descriptive, not causal:** Differences by league or venue may be associated with team strength, tactics, scheduling, and other unobserved factors.
10. **Potential source and parsing errors:** Team labels and event classifications were parsed from source data. Team IDs should be preferred for joins and modeling.

14 13. Recommendations for follow-up analysis

The following extensions would strengthen the analysis:

- **Model goals as a count outcome:** Compare Poisson and negative-binomial models, checking overdispersion and fit.
- **Use a match-aware model:** Include match-level effects or cluster uncertainty by `match_id` because home and away records are paired.
- **Add team strength:** Include team and opponent effects, prior-season performance, or an Elo-style rating to separate venue effects from team quality.
- **Improve attacking measures:** Add expected goals, shot locations, shot type, and goalkeeper context if available.
- **Validate event definitions:** Reconcile tagged shots on target and goals with the original event records and provider documentation.
- **Use multiple seasons:** Test whether league and home-advantage patterns are stable over time.
- **Consider rates with denominators:** For example, goals per shot and tagged shots per attempt should be calculated with clear denominator definitions and appropriate handling of zero-shot matches.

15 Appendix A. Full data dictionary

```
knitr::kable(data_dictionary, caption = "Supplied data dictionary")
```

Table 29: Supplied data dictionary

variable	definition	source	notes
match_id	Unique match identifier	Matches	Shared join key
match_date	Match date (UTC)	Matches	2017–18 season

variable	definition	source	notes
league	Domestic competition	ZIP filename	Five leagues
team_id	Unique team identifier	Matches	Use ID for joins/models
team_name	Team display name	Match label	Parsed; validate labels
opponent_id	Opposing team identifier	Matches	From other side
opponent_name	Opposing team display name	Match label	Parsed; validate labels
home	1 = home, 0 = away	Matches	Binary indicator
goals	Official goals scored by team	Matches	Regression response
goals_conceded	Official goals scored by opponent	Matches	For context
shots	Shot, free-kick shot, or penalty events	Events	Attempt count
on_target_tagged	Shots with accuracy tag 1801	Events	Proxy; validate against definitions
passes_attempted	Events labeled Pass	Events	Excludes set-piece categories
passes_accurate	Passes tagged accurate (1801)	Events	Numerator
pass_accuracy_pct	100 × accurate passes / attempted passes	Derived	NA if no passes
pass_share_pct	100 × team passes / total match passes	Derived	POSSESSION PROXY, not official possession

16 Appendix B. League-by-league shot audit

```
knitr::kable(shot_audit, caption = "Original shot-type audit data")
```

Table 30: Original shot-type audit data

league	shot_type	attempts	tagged_accurate	tagged_goals	goals_without_accurate_tag
England	Free kick shot	350	100	18	0
England	Open-play shot	8451	2986	914	2
England	Penalty	80	56	56	0
France	Free kick shot	521	163	26	0
France	Open-play shot	8327	2963	873	1
France	Penalty	129	99	99	0
Germany	Free kick shot	299	92	18	0
Germany	Open-play shot	6898	2512	747	1
Germany	Penalty	93	68	68	0
Italy	Free kick shot	415	143	36	0
Italy	Open-play shot	8806	3069	853	0
Italy	Penalty	126	89	89	0
Spain	Free kick shot	453	152	28	0
Spain	Open-play shot	7979	2960	884	0
Spain	Penalty	113	81	81	0

17 Appendix C. Session information

```
sessionInfo()
```

```
R version 4.6.1 (2026-06-24 ucrt)
```

```
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26300)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: America/New_York
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] knitr_1.52      scales_1.4.0    lubridate_1.9.5 forcats_1.0.1
 [5] stringr_1.6.0   dplyr_1.2.1     purrr_1.2.2     readr_2.2.0
 [9] tidyr_1.3.2     tibble_3.3.1    ggplot2_4.0.3   tidyverse_2.0.0

loaded via a namespace (and not attached):
 [1] generics_0.1.4     stringi_1.8.9      lattice_0.22-9     hms_1.1.4
 [5] digest_0.6.39      magrittr_2.0.5     evaluate_1.0.5     grid_4.6.1
 [9] timechange_0.4.0   RColorBrewer_1.1-3 fastmap_1.2.0      Matrix_1.7-5
[13] jsonlite_2.0.0     mgcv_1.9-4         cli_3.6.6          rlang_1.3.0
[17] crayon_1.5.3       bit64_4.8.6        splines_4.6.1      withr_3.0.3
[21] yaml_2.3.12        otel_0.2.0         tools_4.6.1        parallel_4.6.1
[25] tzdb_0.5.0         vctrs_0.7.3        R6_2.6.1           lifecycle_1.0.5
[29] bit_4.6.0          vroom_1.7.1        pkgconfig_2.0.3    pillar_1.11.1
[33] gtable_0.3.6       glue_1.8.1         xfun_0.60          tidyselect_1.2.1
[37] rstudioapi_0.19.0  farver_2.1.2       htmltools_0.5.9    nlme_3.1-169
[41] rmarkdown_2.32     labeling_0.4.3     compiler_4.6.1     S7_0.2.2
```